Paleomagnetism, U–Pb geochronology, and geochemistry of Lac Esprit and other dyke swarms, James Bay area, Quebec, and implications for Paleoproterozoic deformation of the Superior Province[1]

Kenneth L. Buchan, Jean Goutier, Michael A. Hamilton, Richard E. Ernst, and William A. Matthews

Abstract: An extensive set of north- to northwest-trending diabase dykes, termed the Lac Esprit swarm, is identified in the Superior Province east of James Bay based on geological mapping and a distinctive paleomagnetic pole (61.7°N, 169.1°E, dm = 7.7°, dp = 5.5°). The Lac Esprit swarm yields a U–Pb baddeleyite age of 2069 ± 1 Ma similar to that of the 2076^{+5}_{-4} Ma Fort Frances swarm of the western Superior Province. Their paleomagnetic declinations differ by 23° ± 12° after correction to a common reference locality. The difference is likely due mainly to counterclockwise rotation about a vertical axis of the Fort Frances area relative to the Lac Esprit area. Differential rotation of 10°–20° has been proposed more locally across the Kapuskasing Structural Zone separating the eastern and western Superior Province in earlier paleomagnetic studies of ca. 2450 Ma Matachewan and 2170 Ma Biscotasing dyke swarms. Thus, relative rotation may have involved the entire eastern and western Superior Province, perhaps in response to collisional events associated with the Trans-Hudson Orogen to the north or the Penokean orogen to the south, or in response to rifting beneath Hudson Bay. Other dykes in the study area are interpreted from a combination of paleomagnetism, trend, and geochemistry to belong to the Senneterre, Matachewan, and Mistassini swarms. The 2216 Ma Senneterre dykes form part of a giant swarm that fans across the eastern Superior Province. Paleomagnetic directions and geometry of this swarm rule out substantial block rotations within the eastern Superior Province since dyke emplacement.

Résumé : Un vaste ensemble de dykes de diabase, à tendance N à NO, nommé l'essaim de dykes du lac Esprit, est identifié dans Province du Supérieur à l'est de la baie James; cette identification est basée sur une cartographie géologique et un pôle paléomagnétique distinctif (61,7°N, 169,1°E, dm = 7,7°, dp = 5,5°). L'essaim du lac Esprit a donné un âge U–Pb sur baddeleyite de 2069 ± 1 Ma, ce qui ressemble à l'essaim de Fort Frances, 2076^{+5}_{-4} Ma, dans l'ouest de la Province du Supérieur. Leurs déclinaisons paléomagnétiques diffèrent de 23° ± 12°, après une correction à une localité de référence commune. La différence est sans doute due à la rotation, dans le sens antihoraire, autour d'un axe vertical dans le secteur de Fort Frances par rapport au secteur du lac Esprit. Dans des études paléomagnétiques antérieures des essaims de dykes de Matachewan, 2450 Ma, et de Biscotasing, 2170 Ma, une rotation différentielle de 10°–20° a été proposée plus à proximité, à travers la zone structurale de Kapuskasing qui sépare les parties est et ouest de la Province du Supérieur. Ainsi, la rotation affecterait les parties est et ouest de la Province du Supérieur, peut-être en réponse à des événements de collision associés à l'orogène trans-hudsonien au nord, ou à l'orogène pénokéen vers le sud, ou encore en réponse à une distension sous la baie d'Hudson. À partir d'études de paléomagnétisme, de tendances et de géochimie, d'autres dykes dans le secteur à l'étude appartiendraient aux essaims de Senneterre, de Matachewan et de Mistassini. Les dykes de Senneterre, 2216 Ma, forment une partie d'un gigantesque essaim qui s'étale dans tout

Received 4 August 2006. Accepted 16 November 2006. Published on the NRC Research Press Web site at http://cjes.nrc.ca on 15 June 2007.

Paper handled by Associate Editor W.J. Davis.

K.L. Buchan.[2] Geological Survey of Canada, 601 Booth Street, Ottawa, ON K1A 0E8, Canada.
J. Goutier. Géologie Québec, 82, boulevard Québec, Rouyn-Noranda, QC J9X 6R1, Canada.
M.A. Hamilton. Jack Satterly Geochronology Laboratory, Department of Geology, University of Toronto, Toronto, ON M5S 3B1, Canada.
R.E. Ernst. Ernst Geosciences, 43 Margrave Avenue, Ottawa, ON K1T 3Y2, Canada, and Department of Earth Sciences, Carleton University, Ottawa, ON K1S 5B6, Canada.
W.A. Matthews. 18 East Sterling Pond Circle, The Woodland, TX 77382, USA.

[1]Geological Survey of Canada Contribution 20060062 and Ministère des Ressources naturelles et de la Faune du Québec Contribution 2005-5130-05.
[2]Corresponding author (e-mail:kbuchan@nrcan.gc.ca).

 doi:10.1139/E06-131 © 2007 NRC Canada

l'est de la Province du Supérieur. Les directions paléomagnétiques et la géométrie de cet essaim ne permettent pas de grandes rotations de blocs dans la partie est de la Province du Supérieur depuis la mise en place des dykes.

[Traduit par la Rédaction]

Introduction

During the past four decades, the study of Paleoproterozoic diabase dyke swarms of the southern Superior Province (Fig. 1) has provided insights into a variety of important topics, including mechanisms of dyke swarm emplacement, plate tectonic processes, continental drift, crustal-scale deformation, and mantle plumes (e.g., Fahrig et al. 1965, 1986; Halls 1986; Bates and Halls 1990; Osmani 1991; Buchan et al. 1993, 1996; Heaman 1997; Halls and Davis 2004; Halls et al. 2005). The age, trend, petrography, geochemistry, and paleomagnetism of most of the larger swarms, such as the 2473–2446 Ma Matachewan, 2216 Ma Senneterre, 2170–2167 Ma Biscotasing, 2126–2101 Ma Marathon, and 2076 Ma Fort Frances swarms, are now well established. However, the overall extent of each has yet to be fully delineated.

In the northeastern Superior Province, between Hudson and Ungava bays, a limited amount of paleomagnetic and geochronological work has also been carried out on the 2230–2210 Ma Maguire and Klotz dyke swarms and the 1998 Ma Minto swarm (Fig. 1; Buchan et al. 1998). Between the southern and northeastern Superior Province, east of James Bay, little systematic study of dyke swarms had taken place prior to the present work, except for preliminary paleomagnetic analysis of six dykes near the La Grande River (Seguin et al. 1983). As a result, dykes east of James Bay have not previously been assigned to swarms.

Geochronological, paleomagnetic, and geochemical studies of dykes east of James Bay are important for defining the extent of the large swarms that have been identified both to the south and north. In turn, this will allow better understanding of the overall geometry of these swarms (radiating or linear), their origin (possible relationship to mantle plumes), and their tectonic significance (possible association with cratonic margins and continental breakup). In addition, the overall geometry and primary paleomagnetic direction for a swarm can be used to detect block rotations that have occurred since swarm emplacement.

During the course of detailed geological field mapping by Géologie Québec (part of Ministère des Ressources naturelles et de la Faune du Québec) in the region east of James Bay (Figs. 2, 3; e.g., Goutier et al. 1998*a*, 1998*b*, 1999*a*, 1999*b*, 2000, 2001*a*, 2001*b*, 2002), diabase dykes have been systematically mapped and material collected for paleomagnetic, geochronological, and geochemical analysis. Many dykes in the area trend N to NW. As discussed in this paper, most dykes of this trend carry an up SSE paleomagnetic direction that has not previously been observed in dykes of the eastern Superior Province. These dykes, referred to as Lac Esprit dykes (Goutier et al. 1999*b*), are the main focus of this paper. In addition, north- to northwest-trending dykes that are thought to belong to other swarms, as well as several prominent northeast-trending dykes that traverse the region, are discussed.

Geological setting

The study area in the region of the Robert-Bourassa reservoir (Fig. 3) is composed largely of 3.36–2.6 Ga Archean rocks, which are crosscut by numerous gabbroic dyke swarms of Proterozoic age and overlain by Proterozoic arenitic rocks of the Sakami Formation in fault-bounded basins (Goutier et al. 1998*a*, 1998*b*, 1999*a*, 1999*b*, 2000, 2001*a*, 2001*b*, 2002). The Archean rocks are typically volcanic–sedimentary assemblages that have been metamorphosed to greenschist or amphibolite facies. They are variably unconformable on basement gneisses and are cut by numerous phases of Archean granitoids.

In the study area of Fig. 3, five Proterozoic dyke swarms are distinguished by their orientations, paleomagnetic characteristics, aeromagnetic signatures, and geochemistry. Individual dykes can commonly be traced for great distances with the assistance of air photographs and aeromagnetic maps. They have an average width of ~35 m and a maximum width of 100 m. They are little metamorphosed or unmetamorphosed. Primary minerals are fine to medium grained and exhibit an intergranular texture (some swarms include dykes that are plagioclase porphyritic as discussed in the following text). Augite is fresh or locally uralitized, whereas plagioclase is commonly saussuritized. Granular magnetite is very common. Quartz is observed only in certain dykes. In outcrop, the Proterozoic dykes are readily distinguished from Archean gabbros because of their brown weathering patina, fresh grey-green fractures, sharp and steep (>71°) intrusive contacts with Archean host rocks, and narrow chilled margins.

Lac Esprit dykes (labelled LE), which trend north to northwest, are the most numerous in the sampling area. They are distinguished from other dykes of similar trends based mainly on paleomagnetic characteristics. They have a relatively weak aeromagnetic signature. Some are aphyric, whereas others vary from porphyritic to glomeroporphyritic with plagioclase. The plagioclase phenocrysts constitute 1%–5% of the dykes by volume and range in size from 0.5 to 2 cm.

In the field, north-trending Matachewan (MT) and NNW- to northwest-trending Lac Shpogan (LS) dykes are not distinguished from Lac Esprit dykes of similar trend. Matachewan dykes are differentiated in the laboratory on the basis of paleomagnetism and geochemistry. Based on interpretation of aeromagnetic data, Matachewan dyke MT1 (Fig. 3) appears to continue south for 500 km (Figs. 1, 2) and represents the most easterly dyke yet identified in the widespread Matachewan swarm. It should be noted that dykes of the Matachewan swarm outside the study area are commonly plagioclase porphyritic. Lac Shpogan dykes are differentiated solely on the basis of paleomagnetism.

Mistassini (MS) dykes trend northwest. Dyke MS3 can be traced 285 km from the study area (Fig. 3) to Lac Mistassini (Fig. 2), where the swarm was originally described (e.g., Fahrig et al. 1986). Mistassini dykes are distinguished from

© 2007 NRC Canada

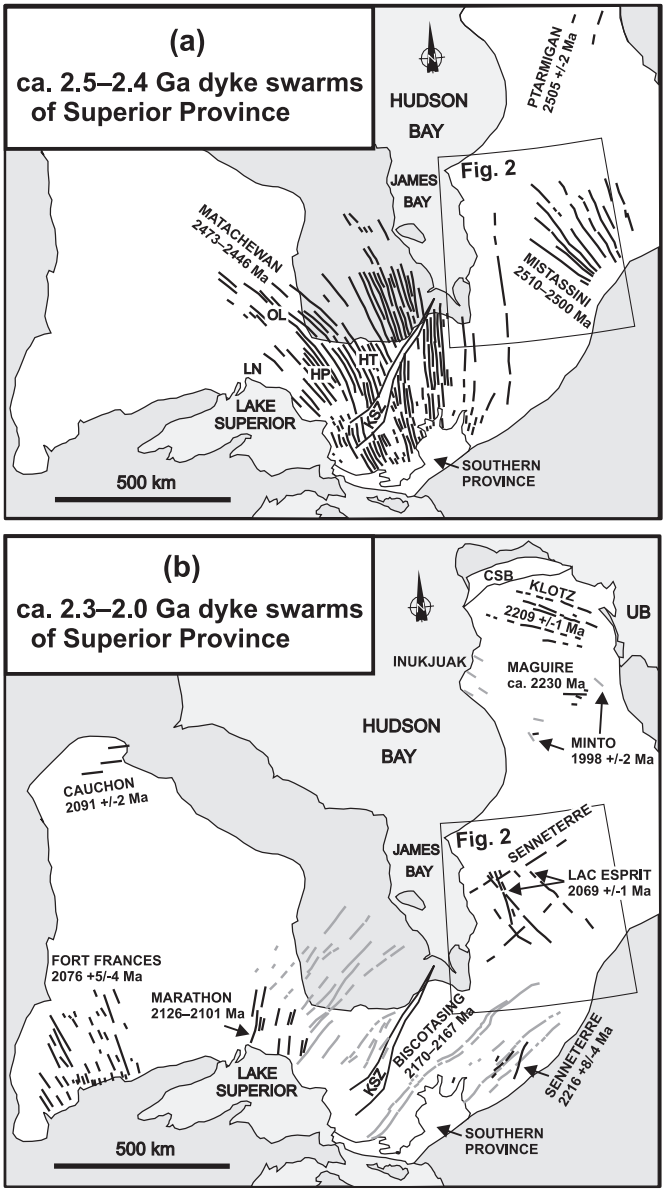


Fig. 1. Selected Paleoproterozoic diabase dyke swarms of Superior Province. KSZ, CSB, and UB are Kapuskasing Structural Zone, Cape Smith Belt, and Ungava Bay, respectively. HP, HT, LN and OL locate the Hornepayne, Hearst, Lake Nipigon, and Ogoki Lake areas, respectively. Dykes of the Biscotasing, Minto, and Inukjuak swarms are shown in grey to distinguish them from other nearby swarms.

other dykes in the region by a pronounced magnetic signature, distinct geochemistry, and the presence of interstitial quartz, microcline, apatite, and rutile commonly forming a microgranophyric texture. In some dykes, a porphyritic texture with large idiomorphic crystals of plagioclase (1–8 cm in length) is observed.

Senneterre (SE) dykes are readily distinguished from the north- to northwest-trending swarms on the basis of their northeast to ENE trend (Figs. 2, 3), paleomagnetic signature, and geochemistry. They are distinguished from other northeast-trending dykes of the Biscotasing and Abitibi swarms (Fig. 2) based on paleomagnetism. They are correlated with previously identified Senneterre dykes to the south and with Maguire and Klotz dykes to the north (Fig. 1*b*), as discussed later in the text. They have elevated MgO contents (9%–10%) compared with other swarms in the study area. Some small minerals, possibly olivine, are completely replaced by chlorite and serpentine.

The Sakami Formation (Fig. 3), composed principally of clastic sedimentary rocks, was defined by Eade (1966). Its dominant unit consists of thick beds of quartz arenite. The Sakami Formation occurs in the interior of basins that extend for 300 km along the original course of the La Grande River. Flooding caused by hydroelectric development along the river has submerged much of the formation (Figs. 2, 3), although it forms islands within the reservoirs (Fig. 4). The Sakami Formation, whose strata dip gently (<30°) to the SSE and W, lies unconformably on Archean rocks and is bounded by normal faults.

Sampling and laboratory methods

Paleomagnetism

During geological field work, approximately five paleomagnetic hand samples were oriented with magnetic compass at each of 38 dyke sites (sites 01–38 of Figs. 2, 3). At three additional sites shown in Figs. 2 and 3 (sites 39–41), paleomagnetic samples were not obtained. Thirty-one paleomagnetic sites are in north- to northwest-trending dykes. The remaining seven sites are in northeast-trending dykes.

In the laboratory, cylindrical specimens with a diameter of 2.5 cm and length of 2.2 cm were drilled from each sample. Their remanent magnetization was measured using a Schonstedt spinner magnetometer. Specimens were subjected to stepwise alternating field (AF) demagnetization, thermal demagnetization, or a combination of AF demagnetization followed by thermal demagnetization. All experiments were carried out within a shielded room with a residual magnetic field <3000 nT, located at the Geological Survey of Canada (GSC) in Ottawa, Ontario. Paleomagnetic directions for individual specimens were determined using Kirschvink (1980) analysis. Site mean directions were determined from specimen directions or, in two cases, from remagnetization circle analysis (Bailey and Halls 1984).

Geochronology

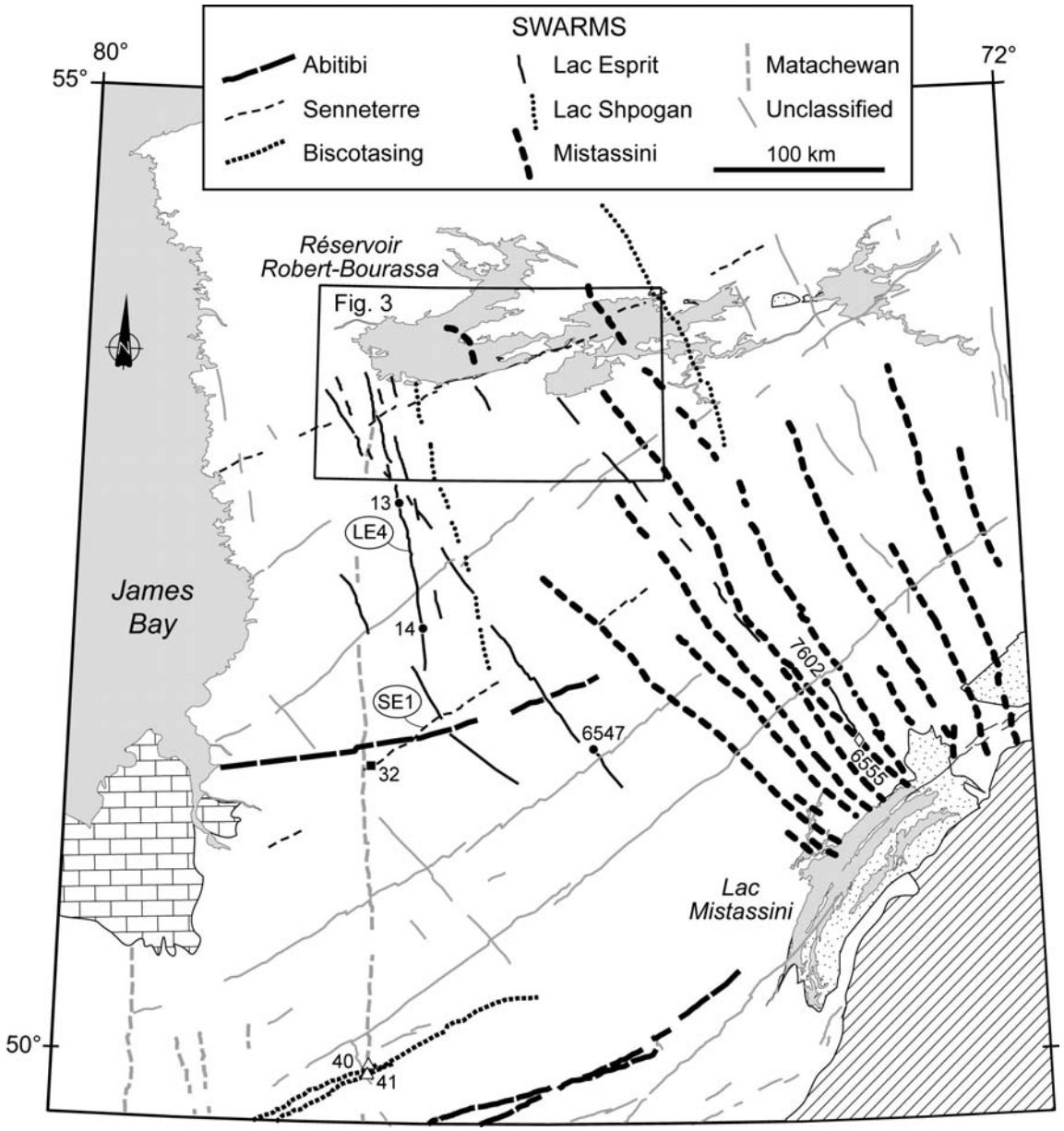
Samples were collected for possible U–Pb geochronological study from the coarse-grained interiors of dykes at several paleomagnetic sites. Based on the paleomagnetic results described in this paper, geochronological samples were selected for analysis from three sites: site 11 in north- to NNW-trending Lac Esprit dyke LE4, site 22 in northwest-trending Lac Esprit dyke LE7, and site 20 in NNW-trending Lac Shpogan dyke LS1 (Fig. 3). U–Pb analytical work was carried out in the Geochronology Section at the GSC. Sample preparation methods are described by Hamilton et al. (2001), along with data from site 11.

Geochemistry

Major and trace elements were analyzed at the GSC Analytical Laboratory. The only exceptions were samples from

© 2007 NRC Canada

Fig. 2. Paleoproterozoic dykes swarms east of James Bay, Quebec (from Buchan and Ernst 2004, with updates using geological and aeromagnetic maps of Système d'information géominière du Québec). The main portion of the sampling area (box) is enlarged in Fig. 3. Lac Esprit paleomagnetic sites 13 and 14, Matachewan paleomagnetic site 32, Matachewan geochemical sites 40 and 41, and sites 6547, 6555, and 7602 of Fahrig et al. (1986) that fall outside the main sampling area are shown. The Grenville Province, Proterozoic sedimentary rocks, and Paleozoic sedimentary rocks are shown with lined, dotted, and brick patterns, respectively. Sampling sites are coded as follows: Lac Esprit (circles), Mistassini (diamonds), Matachewan (triangles), Senneterre (squares).



sites 40 and 41 (Fig. 2) on the presumed southward continuation of Matachewan dyke MT1 that were analyzed at ACME laboratories, Vancouver, British Columbia.

Paleomagnetic results and interpretation

Most paleomagnetic sites yield stable and internally consistent directions of remanent magnetization during AF or thermal demagnetization experiments. Mean remanence directions and statistical parameters for dykes and sites are listed in Tables 1, 2, and 3 and plotted in Fig. 5. In some cases, two or more sites are interpreted to lie along a single dyke on the basis of geological or aeromagnetic continuity, or on the basis of dyke segments that are on strike with one another.

North- to northwest-trending dykes

Nineteen sites in eight north- to northwest-trending dykes (sites 01–04, 07, 08, 10–14, 16–18, 22–24, 27, 28) are stably magnetized with an intermediate up remanence directed to the SSE (Table 1; Fig. 5*a*). In general, directions from sites collected along a single dyke are more tightly clustered than those from different dykes. Examples of AF and thermal demagnetization experiments are shown in Figs. 6 and 7,

© 2007 NRC Canada

Fig. 3. Sampling sites and dykes of the main sampling area. Sakami Formation sedimentary rocks are shown in a dotted pattern, including those submerged within the Robert-Bourrassa and LG-Trois reservoirs. Dykes are coded as in the legend of Fig. 2. Sampling sites are coded as follows: Lac Esprit (circles), Lac Shpogan (pentagons), Mistassini (diamonds), Matachewan (triangles), Senneterre (squares), unclassified (cross). Sites outside the main sampling area are shown on Fig. 2.

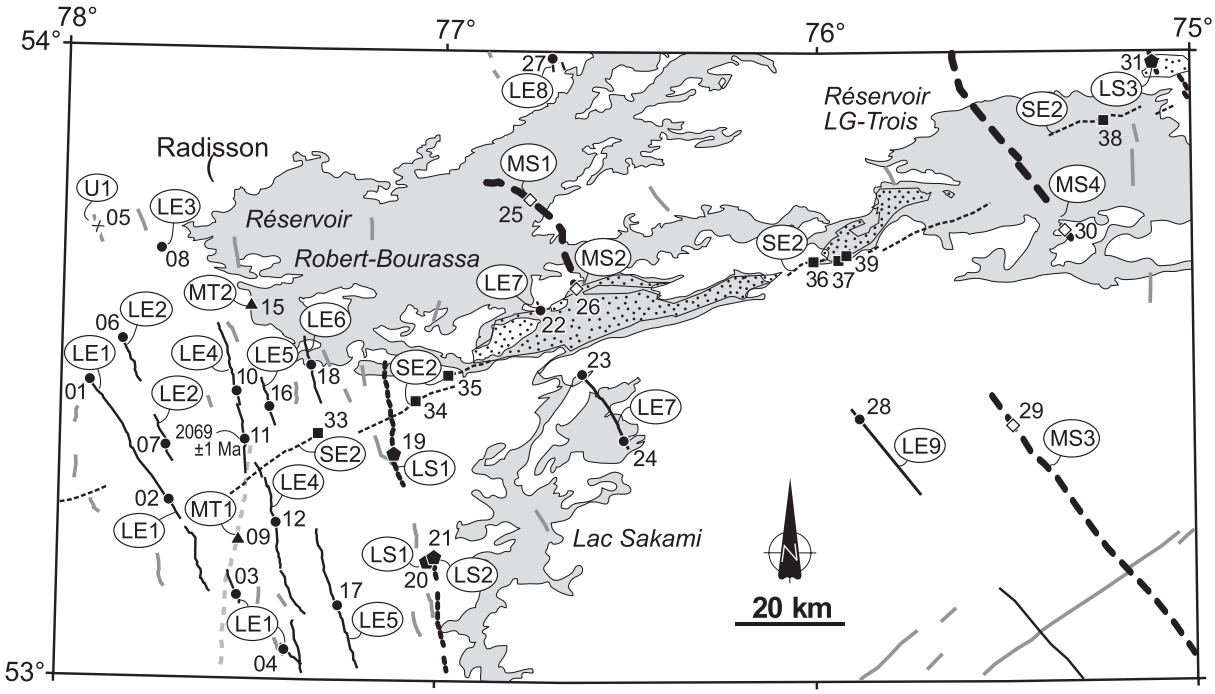
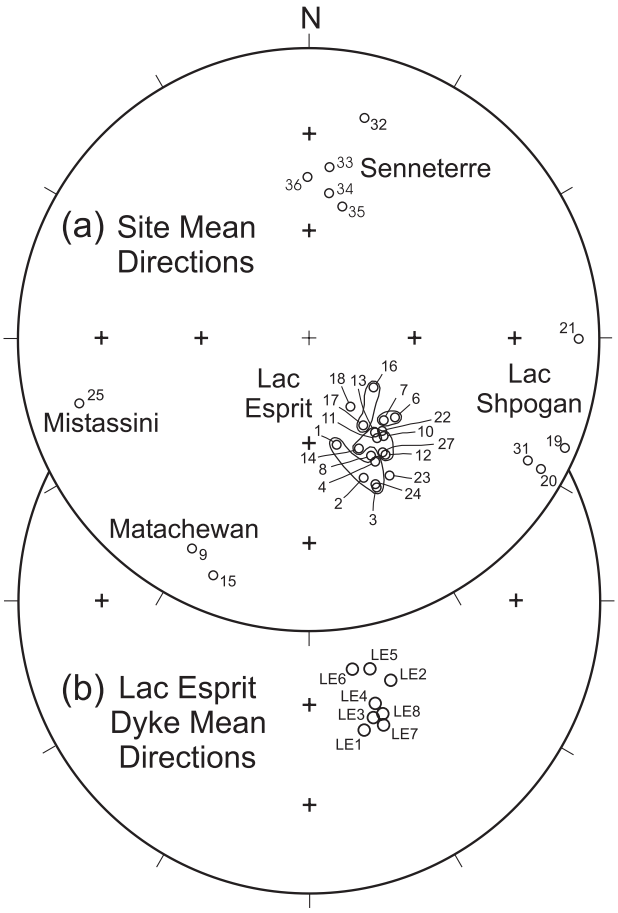


Fig. 4. Quartz arenite of the Sakami Formation, with beds dipping shallowly to the SSE and cut by a steeply dipping, 30 m wide, dyke of the Lac Esprit swarm (site 22). View towards the southeast on Robert-Bourassa reservoir.



© 2007 NRC Canada

Fig. 5. Mean paleomagnetic directions (corrected to a common reference locality at 53.5°N, 77.0°W). Data are plotted on equal area nets with open symbols indicating up directions. (*a*) Site mean directions. For Lac Esprit dykes, a closed curve is drawn around directions that are interpreted to be from sites along a single dyke. For clarity the curve is omitted for sites 22, 23 and 24 of dyke LE7. Possible secondary overprints from sites 05 and 26 are not shown. (*b*) Lac Esprit dyke mean directions.



respectively. Typically a small component, probably of viscous origin, is eliminated in the first few demagnetization steps, before the stable remanence is isolated. Blocking temperatures of this stable component are generally concentrated between 450 and 575 °C (Fig. 7), indicating that it is carried mostly by low-titanium titanomagnetite.

The dykes that carry the up SSE remanence (Table 1) are referred to collectively as Lac Esprit dykes (Goutier et al. 1999*b*). Paleomagnetic data at site 28 in northwest-trending dyke LE9 (Fig. 3) suggest that it also belongs to the Lac Esprit swarm, although the data were not of sufficient quality to allow their use in statistical calculations.

At four Lac Esprit sites (sites 06, 08, 22, and 24), there is evidence in a few samples during thermal or AF demagnetization of a superimposed magnetization component roughly antipodal to the dominant SSE component. A particularly dramatic example is illustrated in Fig. 6*b*. The explanation of such antipodal magnetizations is uncertain. Dykes are thought to cool much too rapidly through their blocking temperature range to acquire thermoremanent magnetizations of two polarities during a field reversal. Blocking temperature spectra of the antipodal components overlap significantly, ruling out a partial thermal or viscous-thermal overprint. The down NNW magnetization may be a chemical overprint of unknown origin, acquired after a field reversal that followed acquisition of a primary up SSE thermoremanent magnetization. Such dual polarity results in individual dykes are observed in a small number of other swarms, such as the Matachewan (Halls and Zhang 2003) and Marathon swarms (Buchan et al. 1996) of the southern Superior Province.

Mean paleomagnetic directions were calculated for each of eight Lac Esprit dykes (Table 2; Fig. 5*b*). They form a tight grouping with an overall mean direction of $D = 147.4°$, $I = –55.9°$ ($k = 106.6$, $\alpha_{95} = 5.4°$) and a corresponding paleomagnetic pole at 61.7°N, 169.1°E (dm = 7.7°, dp = 5.5° (semi-axes of the oval of confidence about the mean pole)). A similar paleopole at 61.9°N, 175.3°E (dm = 9.0°, dp = 6.6°) is obtained using only the six north- to NNW-trending dykes (Table 2).

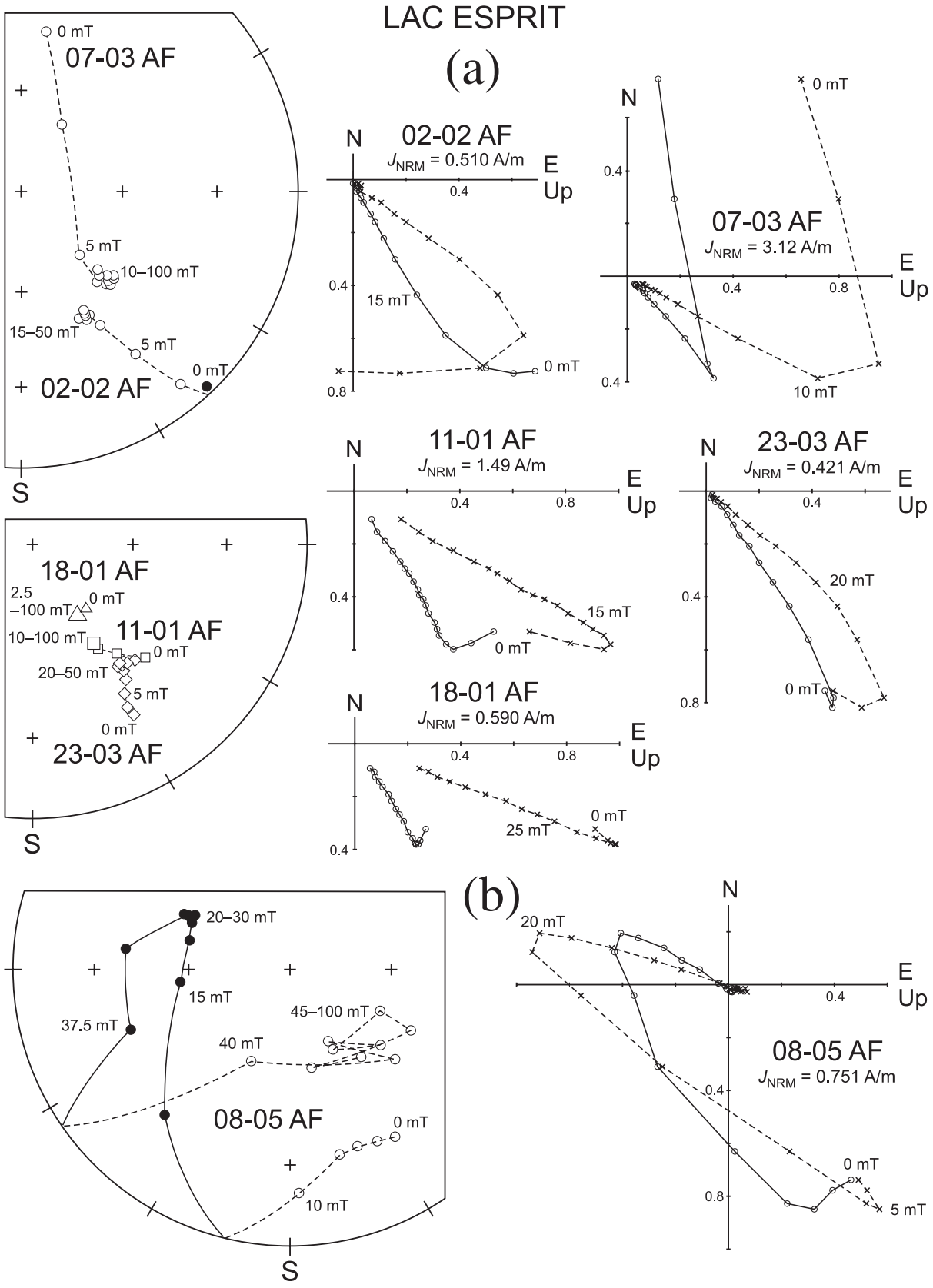
Eleven sites in north- to northwest-trending dykes do not carry the up SSE Lac Esprit magnetization (Table 1). Sites 09 and 15 have shallow up SSW remanences (Figs. 5*a*, 8*a*), indicating that these dykes belong to the ~2450 Ma Matachewan swarm. As noted earlier in the text, dyke MT1 (site 09) may represent the extension of a Matachewan dyke that can be followed on the basis of its aeromagnetic signature for 500 km to the south (Fig. 2).

Four sites (sites 19, 20, 21, and 31) in north- to NNW-trending dykes yield shallow ESE to east magnetization directions (Figs. 5*a*, 8*b*) with a mean of $D = 110.1°$, $I = –10.0°$ ($k = 31.4$, $\alpha_{95} = 16.7°$). Blocking temperatures of the stable magnetization are 450–575 °C, indicating that it is carried by low-titanium titanomagnetite. These dykes have not been correlated conclusively with any specific swarm on the basis of paleomagnetism or geochemistry (see later in the text) and are tentatively referred to as Lac Shpogan dykes.

Four NNW- to northwest-trending dykes (sites 25, 26, 29, and 30) are interpreted as belonging to the 2510–2500 Ma (Heaman 2004) Mistassini swarm, which was studied in detail by Fahrig et al. (1986) in the vicinity of Lac Mistassini (Fig. 2), 200 km to the southeast of the sampling area. This correlation is based on an analysis of trends, aeromagnetic continuity, geochemistry, and paleomagnetism. As described in the section on "Geochemistry," all four dykes have geochemical signatures similar to those of the Mistassini swarm, yet unlike those of any other swarm in the area. One dyke (site 29 of Fig. 3) can be traced on aeromagnetic maps for 285 km to the southeast and represents the continuation of a prominent Mistassini dyke studied by Fahrig et al. at their sites 6555 (Fig. 2) and 7603 (not shown). All four sites have complicated paleomagnetic signatures, as observed by Fahrig et al. in Mistassini dykes to the southeast (Fig. 8*c*). Fahrig et al. (1986) found that Mistassini dykes carry three magnetization components that are directed steeply down (or up), northeast and down (or southwest and up), and northeast and up (or southwest and down). In some instances, they also observed intermediate directions and swings indicating superposition of two or more components. In this study, the four Mistassini sites also yielded rather complex multicomponent paleomagnetic results (Table 1; Fig. 8*c*). Site 29 shows long swings in remanence on AF demagnetization towards a poorly defined

© 2007 NRC Canada

Fig. 6. (*a*) Paleomagnetic data for north- to northwest-trending Lac Esprit dykes on alternating field demagnetization. (*b*) Dual polarity remanence from a Lac Esprit dyke. Closed (open) symbols on equal area projections indicate down (up) directions. Circles (crosses) on orthogonal component diagrams represent projection onto the horizontal (vertical) plane. J_{NRM} is the intensity of natural remanent magnetization prior to demagnetization.



© 2007 NRC Canada

Table 1. Summary of paleomagnetic results for north- to northwest-trending dykes.

Site	Dyke	Site location (°N, °W) NAD83	Trend at site (°)	Dyke trend	Width at site (m)	*N*, *n*	*D*(°), *I*(°)	*D*′(°), *I*′(°)	α_{95} (°)	*k*
Lac Esprit dykes										
01 (CB97-6387)	LE1[a]	53.470, 77.926	312	NW	30	5, 5	164.5, –59.2	164.8, –59.1	10.3	55.9
02 (SC97-8029)	LE1[a]	53.281, 77.709	335	NW	60	5, 5	158.1, –47.1	158.4, –47.2	10.2	57.3
03 (MH97-7264)	LE1[a]	53.133, 77.526	327	NW/NNW	85	5, 4	155.6, –42.8	155.8, –43.0	12.5	54.7
04 (MH97-7069)	LE1[a]	53.047, 77.398	318	NW	35	5, 4	151.1, –50.2	151.2, –50.5	16.6	31.7
06 (JG97-1864)	LE2	53.536, 77.841	332	NNW	23	6, 6	132.1, –56.8	132.5, –56.4	7.1	89.4
07 (JG99-001)	LE2	53.369, 77.721	337	NNW	33	5, 5	137.3, –58.8	137.5, –58.6	8.1	90.6
08 (NR97-737)	LE3	53.680, 77.743	005	N	15	5, 4	151.5, –52.5	151.9, –52.1	10.1	83.6
10 (JG97-1748)	LE4	53.456, 77.535	333	NNW	60	5, 5	142.3, –55.0	142.5, –54.9	4.5	293.7
11 (JG97-999)[b]	LE4	53.380, 77.511	000	N/NNW	45	5, 5	145.5, –55.9	145.7, –55.8	7.2	113.1
12 (MH97-7196)	LE4	53.249, 77.427	347	NNW	50	5, 5	146.5, –50.3	146.6, –50.4	9.5	66.4
13 (KM99-480)	LE4	52.870, 77.283	357	N/NNW	50	5, 5	144.9, –57.0	144.8, –57.4	7.3/12.0[c]	330.0
14 (KM99-386)	LE4	52.269, 77.080	001	N/NNW	17	5, 4	156.0, –54.4	155.7, –55.5	13.4	47.7
16 (JG99-006)	LE5	53.432, 77.448	015	NNW	30	5, 5	126.6, –67.1	126.7, –66.9	6.6	136.6
17 (JG98-2284B)	LE5	53.118, 77.260	310	NNW	35	5, 4	147.7, –60.4	147.6, –60.6	7.9	136.7
18 (MH97-7261)	LE6	53.499, 77.339	335	NNW	25	5, 5	148.1, –67.3	148.2, –67.2	7.8	97.9
22 (JG98-2280)[d]	LE7	53.590, 76.734	311	NW	30	5, 5	141.9, –56.8	141.8, –56.8	9.1	72.4
23 (FF98-11066)	LE7	53.489, 76.621	325	NW	30	5, 5	149.8, –44.6	149.6, –44.8	4.8	260.6
24 (JG98-2122)	LE7	53.384, 76.509	330	NW	40	5, 3	155.7, –43.6	155.4, –43.9	9.3	175.2
27 (JG99-1177)	LE8	53.990, 76.709	350	N	1.5	5, 5	147.4, –51.3	147.4, –51.0	7.5	105.0
28 (JG00-1654)	LE9	53.418, 75.890	?	NW	45	5, 0	—	—	—	—
Matachewan dykes										
09 (MH97-7039)	MT1	53.220, 77.524	341	N/NNE	7	5, 3	208.7,–19.6	209.1, –20.3	10.3	143.1
40 (JG02-1243)[e]	MT1	49.956, 77.461	360	N	60	0, 0	—	—	—	—
41 (MC02-5335)[e]	MT1	49.904, 77.467	?	N	50	0, 0	—	—	—	—
15 (CD96-5080)	MT2	53.594, 77.500	344	NNW?	3	5, 5	201.4, –14.5	201.7, –14.6	17.7	19.5
Lac Shpogan dykes										
19 (MH98-4282)	LS1	53.359, 77.117	013	NNW	40	4, 4	112.7, –5.6	112.8, –5.5	10.6	75.7
20 (JG97-1603)	LS1	53.189, 77.027	333	NNW	60	5, 4	118.8, –10.0	118.8, –10.3	14.4	41.6
21 (JG97-1609)	LS2	53.196, 77.009	350	NNW	80	5, 4	89.4, –8.0	89.4, –8.0	21, 28[c]	28.0
31 (JG00-1650)[d]	LS3	53.986, 75.101	341	NNW	60	5, 4	120.5, –14.4	119.1, –15.8	18.1	26.7
Mistassini dykes										
25 (JG99-1190)	MS1[f]	53.764, 76.757	295	NW	23	5, 3	254.7, –20.0	254.5, –19.6	19.0	43.2
26 (JG98-2101)[g]	MS2[f]	53.625, 76.641	329	NNW	70	5, 4	1.5, 69.4	1.4, 69.3	10.2	82.7
29 (JG00-1655)	MS3[f,h]	53.411, 75.483	322	NW	20	4, 0	—	—	—	—
30 (JG00-1652)	MS4[f]	53.722, 75.346	330	NW	4	5, 0	—	—	—	—
Unclassified dyke										
05 (IL97-776)	U1	53.712, 77.916	352	NNW	15	5, 4	277.8, 52.4	278.6, 51.8	13.0	50.6

© 2007 NRC Canada

Table 1 (*concluded*).

Means:
Lac Esprit: See means in Table 2
Lac Shpogan (four sites): Direction is D = 110.1°, I = –10.0° (k = 31.4, α_{95} = 16.7°); virtual geomagnetic pole is 15.9°N, 179.7°E (dm = 16.9°, dp = 8.5°).
Matachewan (two sites): Direction is D = 205.3°, I = –17.5° (angular separation, θ = 9.1°); virtual geomagnetic pole is 41.0°S, 111.1°W.

Note: Dyke are identified on the basis of geological or aeromagnetic continuity or by their probable extension based on location and orientation. They are named according to the swarm with which they are thought to be associated (LE, Lac Esprit; MT, Matachewan; LS, Lac Shpogan; MS, Mistassini; U, unclassified). *N*, number of samples measured in AF demagnetization experiments; *n*, number of samples used in statistical calculation; *D* and *I*, declination and inclination of remanent magnetization direction calculated as the Fisherian mean (Fisher 1953) of sample directions, except for site 13, where the direction is calculated from the intersection of five remagnetization circles following the method of Bailey and Halls (1984), and site 21, where the direction is calculated using two remagnetization circles and two stable endpoints; *D′* and *I′*, declination and inclination recalculated for a common reference locality at 53.5°N, 77.0°W; α_{95}, radius of the circle of 95% confidence about the magnetization direction; *k*, precision parameter (Fisher 1953); dm and dp, semi-axes of the oval of confidence about the mean paleopole.
[a]Dyke LE1 can be traced aeromagnetically southeast into Fahrig et al. (1986) dyke site 6547.
[b]Site that is dated by the U–Pb technique at 2069 ± 1 Ma (see text; aslo see Hamilton et al. (2001) where site is labelled J97-EKN).
[c]Minimum and maximum values of α_{95} calculated from remagnetization circle analysis of Bailey and Halls (1984).
[d]Dykes that crosscut the Sakami Formation (see text).
[e]Geochemical sampling sites; no paleomagnetic samples collected.
[f]Dykes MS1–MS4 have geochemistry similar to dykes of the Mistassini swarm (see text).
[g]Dyke that is overlain by the Sakami Formation (see text).
[h]Dyke MS3 can be traced aeromagnetically southeast into Fahrig et al. (1986) dyke site 6555.

shallow northeast direction. Most samples at site 25 have an up WSW remanence. Site 26 carries a steep down N magnetization close to the present Earth's field that may represent a recent overprint. Finally, site 30 has rather scattered magnetizations, with some samples showing swings towards a shallow up NE direction during demagnetization. Although these paleomagnetic results are of only limited use in establishing the link with the Mistassini swarm, they are not inconsistent with such a correlation.

Finally, site 05 from a north-trending dyke carries a down W remanence reminiscent of that in Biscotasing dykes. However, the Biscotasing swarm has a northeast trend, so that the age of the dyke at site 05 is uncertain. A similar down W remanence is observed as a secondary overprint at Lac Esprit dyke site 08.

Northeast-trending dykes

Five of the seven paleomagnetic sites collected from northeast-trending dykes (Figs. 2, 3) are stably magnetized in a northerly direction with an intermediate negative inclination (Table 3; Fig. 9). Their mean remanence direction is D = 8.7°, I = –41.4° (k = 45.0, α_{95} = 11.5°). The stably magnetized sites include site 32 from dyke SE1 in the southern part of the sampling area, and sites 33, 34, 35, and 38, which appear to lie along a single dyke (SE2) in the northern portion of the area. Thermal demagnetization experiments (Fig. 9*b*) indicate that the stable remanence is carried by low-titanium titanomagnetite with maximum blocking temperatures of 575 °C, with the exception of site 32 in dyke SE1. At site 32, the northerly up component is carried by both low-titanium titanomagnetite and hematite with blocking temperatures up to 685 °C. The hematite may be the result of late stage deuteric alteration.

Dykes SE1 and SE2 have paleomagnetic directions indistinguishable from those of 2216 Ma NNE-trending Senneterre dykes (Buchan et al. 1993) to the south of the sampling area, and hence are considered to form an extension of the Senneterre swarm. Their paleomagnetic directions are also similar to those of ~2230 Ma east-trending Maguire dykes (Buchan et al. 1998) to the north. As discussed in another section, Senneterre and Maguire dykes form part of a giant radiating swarm.

Three additional sites are interpreted to belong to Senneterre dyke SE2. Site 37 has paleomagnetic directions to the north, but with a great deal of scatter; site 36 is unstably magnetized; and site 39 was not studied paleomagnetically. Nevertheless, all of these sites are on dyke segments that are roughly on strike with dyke SE2, and both sites 36 and 37 have geochemistry that is indistinguishable from that of dyke SE2 (see Geochemistry section).

U–Pb geochronology

One of the three sites that were selected for geochronological study yielded baddeleyite (site 11), the others only poor quality zircon (sites 20 and 22).

Site 11 in north- to NNW-trending dyke LE4 on the north shore of Lac Ekomiak is 45 m wide and carries a typical Lac Esprit paleomagnetic direction. U–Pb geochronological results from this site are reported in detail by Hamilton et al. (2001) and summarized briefly here.

© 2007 NRC Canada

Table 2. Mean paleomagnetic results for Lac Esprit dykes and comparison with mean data for Fort Frances dykes.

Dyke	Trend	Nd	Ns	D'(°), I'(°)	α_{95} (°)	k	Plat (°N), Plon (°E)	dm (°), dp (°)
LE1	NW	—	4	157.2, –50.0	8.7	112.6	62.0, 148.2	11.6, 7.8
LE2	NNW	—	2	135.0, –57.5	θ = 3.5	—	55.8, 194.9	—
LE3	N	—	1	151.9, –52.1	—	—	61.2, 158.3	—
LE4	N/NNW	—	5	147.1, –54.9	3.7	430	60.8, 168.2	5.2, 3.7
LE5	NNW	—	2	138.3, –64.1	θ = 11.1	—	62.4, 165.9	—
LE6	NNW	—	1	148.2, –67.2	—	—	70.1, 195.6	—
LE7	NW	—	3	149.6, –48.6	12.9	92.7	57.4, 158.0	16.9, 11.1
LE8	N	—	1	147.4, –51.3	—	—	58.1, 163.1	—
Means:								
all sites	—	—	19	148.3, –54.6	3.5	91.5	61.2, 166.2	5.0, 3.5
all dykes	—	8	—	147.4, –55.9	5.4	106.6	61.7, 169.1	7.7, 5.5
NNW- to N-trending dykes	—	6	—	145.0, –58.0	6.1	121.2	61.9, 175.3	9.0, 6.6
NW-trending dykes	—	2	—	153.4, –49.4	θ = 5.2	—	59.8, 153.3	—
Fort Frances[a]	—	13	—	124.2, –47.4	5.1	68	42.7, 184.2	7.3, 5.2

Note: Nd, number of dykes; Ns, number of sites; D' and I', declination and inclination recalculated for a common reference locality at 53.5°N, 77.0°W; α_{95}, radius of the circle of 95% confidence about the magnetization direction (in cases where Ns = 2 or Nd = 2, the angular difference between the pair of directions is listed); k, precision parameter (Fisher 1953). Plat and Plon, coordinates of the paleomagnetic pole or virtual geomagnetic pole in the northern hemisphere; dm and dp, semi-axes of the oval of 95% confidence about the mean paleopole.

[a]Fort Frances paleomagnetic direction recalculated from Halls (1986) for reference locality 53.5°N, 77.0°W.

Table 3. Summary of paleomagnetic results for northeast-trending (Senneterre) dykes.

Site	Dyke	Site location (°N, °W) NAD83	Trend at site (°)	Dyke trend	Width at site (m)	N, n	D (°), I (°)	D'(°), I'(°)	α_{95} (°)	k
32 (KM98-280)	SE1	51.522, 77.459	058	NE	?	6, 4	13.7, –26.1	14.0, –22.6	24.0	15.7
33 (MH97-7434)	SE2	53.390, 77.318	050	NE	50	5, 4	6.3, –40.5	6.6, –40.4	22.2	18.17
34 (LP96-3364)	SE2	53.444, 77.062	036	NE	60	5, 5	8.1, –48.3	8.2, –48.3	5.1	229.8
35 (MH98-4193)	SE2	53.485, 76.976	075	NE	36	5, 5	14.6, –51.2	14.6, –51.2	5.8	175.0
38 (JG00-1651)	SE2	53.893, 75.233	045	ENE	72	5, 5	1.6, –43.1	359.5, –43.6	14.9	27.4
37 (JG00-1656)	SE2	53.670, 75.938	055	ENE	20	5, 0	—	—	—	—
36 (JG00-1653)	SE2	53.668, 76.003	067	ENE	64	5, 0	—	—	—	—
39 (MH99-4086)[a]	SE2	53.677, 75.928	085	ENE	9	0, 0	—	—	—	—

Mean (five sites):
Direction is D = 8.7° I = 41.4° (k = 50.0, α_{95} = 11.5°); virtual geomagnetic pole is 12.3°S, 85.2°W (dm = 14.1°, dp = 8.6°).

Note: See explanatory notes with Table 1. Sites 33–39 are thought to belong to a single Senneterre dyke (SE2) based on relative locations (Fig. 3), trend, paleomagnetic signature, and (or) geochemistry (see text).

[a]Dyke segment that crosscuts the Sakami Formation; no paleomagnetic samples collected.

Site 11 yielded abundant short, broken blade fragments of fresh, honey-brown baddeleyite. On a U–Pb concordia diagram, data for five baddeleyite fractions are collinear, partly overlapping, and 0.8%–2.0% discordant (Fig. 10, Table 4).

Calculated $^{207}Pb/^{206}Pb$ model ages fall in a range from 2068.3 to 2069.7 Ma (Table 4). The data, when regressed using a modified York (1969) method and anchored assuming modern-day Pb-loss, yield an upper intercept age of 2069 ± 1 Ma (2σ error; mean square of weighted deviates, MSWD = 0.05). This is interpreted to represent the magmatic crystallization age of the dyke.

Zircons from site 20 of NNW-trending dyke LS1 and site 22 of northwest-trending dyke LE7 yielded scattered Archean ages that are interpreted to be of xenocrystic origin.

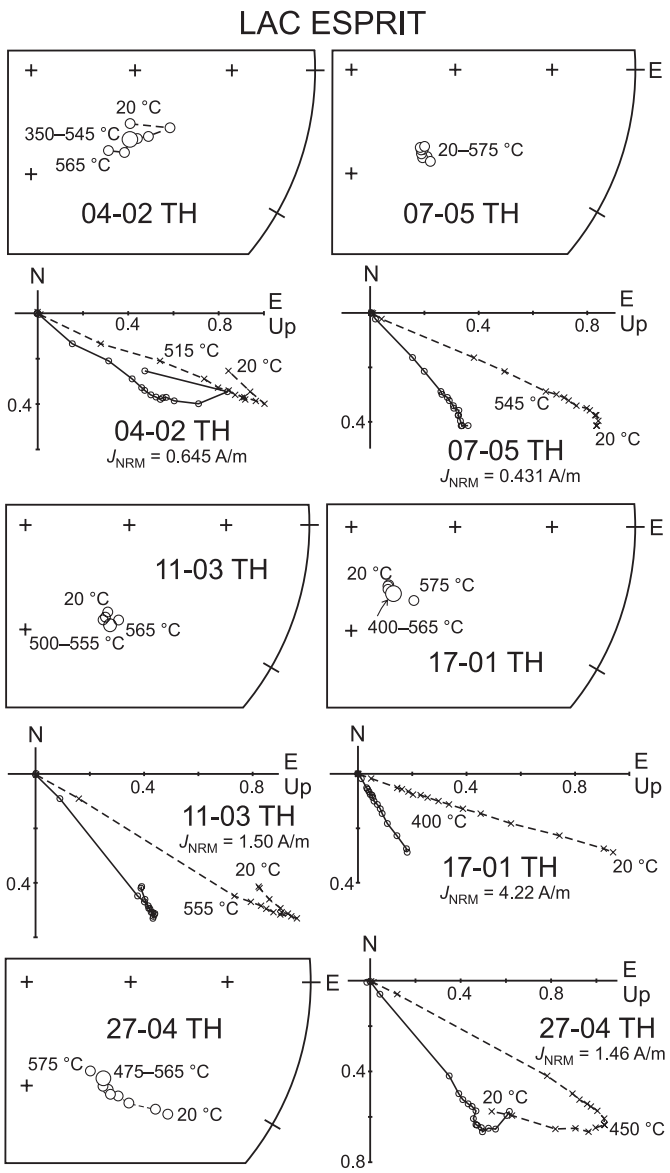
Geochemistry

Geochemical data were obtained during the GSC Targeted Geoscience Initiative project 2001–2003 for each of the dyke swarms of the sampling area (see supplementary data[2]), as well as from reference suites for several dyke swarms across the Canadian Shield (unpublished data of R. Ernst). The distribution of the relevant Paleoproterozoic swarms is shown in Fig. 1. The distribution of the Mesoproterozoic Mackenzie

[2] Supplementary data can be purchased from the Depository of Unpublished Data, CISTI, National Research Council of Canada, Ottawa, ON K1A 0S2, Canada. This will also be available along with the article on the NRC Research Press Web site.

© 2007 NRC Canada

Fig. 7. Paleomagnetic data for north- to northwest-trending Lac Esprit dykes on thermal demagnetization. Symbols as in Fig. 6.



(1267 Ma) and Sudbury (1238 Ma) swarms and the Neoproterozoic Franklin (723 Ma) swarm are shown in Buchan and Ernst (2004).

Based on an AFM (Alkalis–FeO*–MgO) diagram (see supplementary data[3]), most suites from the study area are tholeiites. Senneterre dykes have the most primitive compositions, whereas those correlated with the Mistassini swarm are the most evolved. In addition, north- to NNW-trending Lac Esprit dykes tend to be more evolved than northwest-trending Lac Esprit dykes. On a total alkali vs. silica (TAS) diagram (see supplementary data[3]), most suites plot as basalts, except for interpreted Mistassini dykes, which plot in the trachybasalt and basaltic trachyandesite fields. The single Lac Shpogan sample that was analyzed (from site 31, dyke LS03) plots as a tephrite-basanite close to the basalt field. This diagram also shows that north- to NNW-trending and northwest-trending Lac Esprit dykes plot in distinct fields.

Diagrams of mantle-incompatible trace elements are particularly useful for fingerprinting magmatic events (e.g., Rollinson 1993; Hofmann 1997; Condie 2003). Examples that efficiently separate the present data are shown in Figs. 11–13. Four northwest-trending dykes from the study area (dykes MS1–MS4) have geochemistry identical to most dykes of the Mistassini reference set near Lac Mistassini (Figs. 12, 13) and distinct from the other dyke swarms. The four dykes are interpreted to belong to the Mistassini swarm on the basis of this geochemical correlation as well as other criteria that were discussed earlier. The interpreted southerly extension of Matachewan dyke MT1 (sites 40 and 41 of Fig. 2) has chemistry that is similar to that observed in the main portion of the Matachewan swarm to the west (Fig. 12), but distinct from that of similarly trending north- to NNW-trending Lac Esprit dykes (Fig. 12). The Lac Shpogan dyke is not readily associated with another specific swarm on the basis of geochemistry, as its chemistry is broadly similar to that of the north– to NNW-trending Lac Esprit dykes (Figs. 12, 13), some dykes of the Minto reference suite (Fig. 12), some dykes of the Mackenzie reference suite (Fig. 12), and some dykes of the Sudbury reference suite (Figs. 12, 13). The two Senneterre dykes in the study area differ significantly in geochemistry from each other (Figs. 11–13), but are broadly similar to the Maguire–Senneterre–Nipissing reference data.

In addition, trace element diagrams show remarkable along-dyke geochemical consistency for four of the Lac Esprit dykes and also Senneterre dyke SE2 (Figs. 11, 12*b*, 13*b*). Dyke MS3 in the sampling area of this study can be traced aeromagnetically into and has identical composition to Mistassini dyke 6555 (see Fig. 2) that was studied by Fahrig et al. (1986) in the Lac Mistassini area. Similarly, northwest-trending Lac Esprit dyke LE1 can be traced into and has identical composition to northwest-trending dyke 6547 (Fig. 2), which was originally interpreted as a Mistassini dyke by Fahrig et al. (1986). Finally, the chemistry of northwest-trending Lac Esprit dyke LE9 matches that of northwest-trending dyke 7602 (Fig. 2), also interpreted by Fahrig et al. (1986) as a Mistassini dyke, suggesting a probable link. Therefore, we conclude that dykes 6547 and 7602 belong to the northwest-trending portion of the Lac Esprit swarm rather than the Mistassini swarm.

Discussion

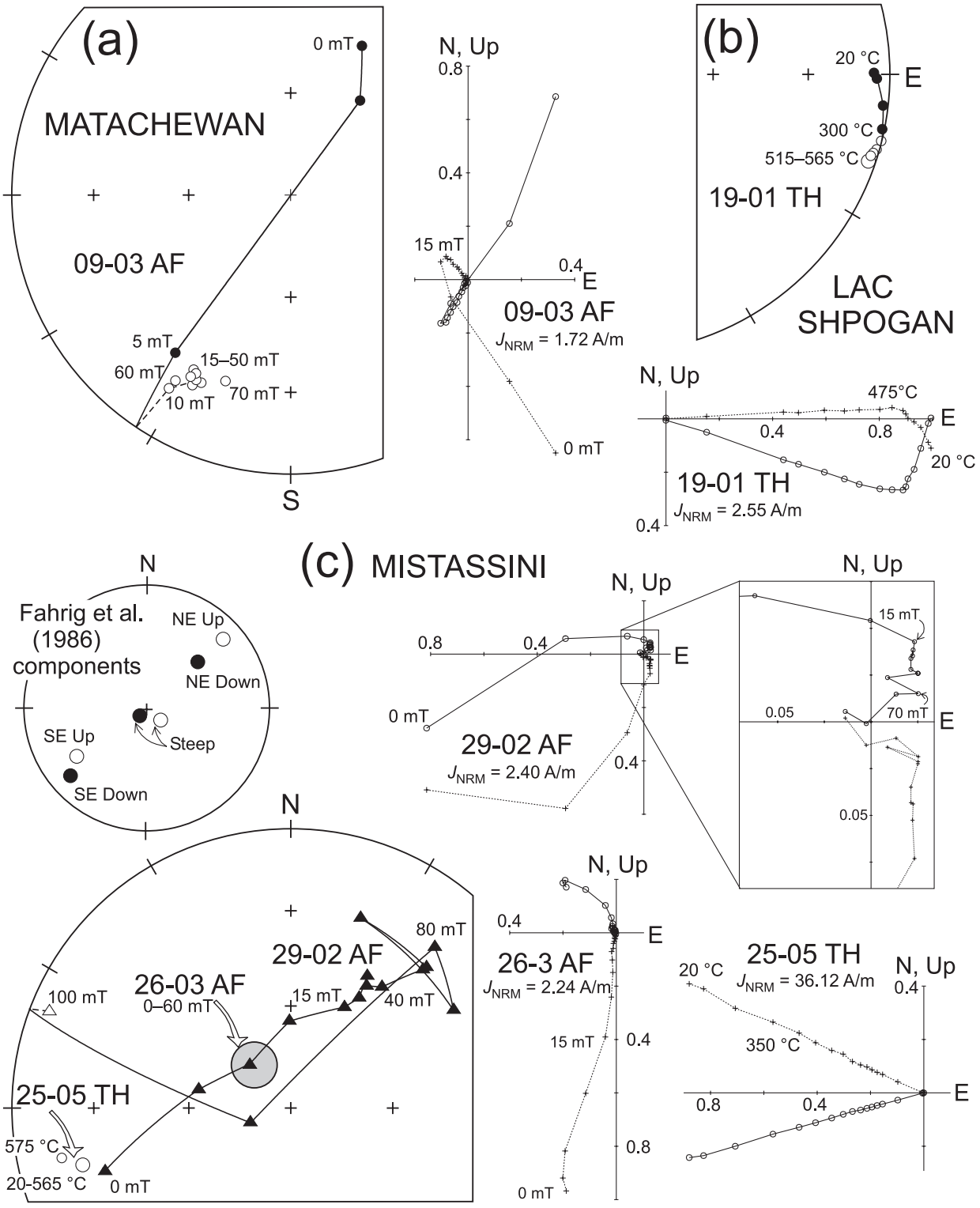
The mean paleopole for Lac Esprit dykes and virtual geomagnetic poles (VGPs) from Senneterre, Matachewan, and Lac Shpogan dykes of this study are plotted with other well-defined Paleoproterozoic poles from the Superior Province in Fig. 14.

Age of magnetic remanence in Senneterre, Lac Esprit, and Mistassini dykes

The paleopole for 2216 Ma Senneterre dykes south of the study area (Fig. 1*b*) is indistinguishable from the N1 paleopole from coeval Nipissing sills of the Southern Province (Fig. 14; Buchan et al. 1993). Positive baked contact tests for the Senneterre (Symons et al. 1981) and N1 Nipissing remanences (Buchan 1991), as well as a positive secular variation test for the Senneterre magnetization (Buchan et al. 1993), demonstrate that the Senneterre–Nipissing remanence is primary. As illustrated in Fig. 14, the VGP for Senneterre

© 2007 NRC Canada

Fig. 8. Paleomagnetic data for north- to northwest-trending dykes that have not been correlated with Lac Esprit dykes. In (*c*), individual data points for sample 26-03 are not shown for clarity but are all directed steeply down and fall within the shaded circle. Similarly, individual data points between 20 °C and 565 °C for sample 25-05 fall within the larger of the pair of open circles. Symbols as in Fig. 6.



dykes in this study (Fig. 5*a*) is indistinguishable from the paleopoles for more southerly Senneterre dykes, N1 Nipissing sills (Buchan et al. 1993), and more northerly Maguire dykes (Fig. 1*b*; Buchan et al. 1996), and therefore is considered to be primary.

Although no baked contact test was carried out in this

© 2007 NRC Canada

Fig. 9. Paleomagnetic data for northeast-trending Senneterre dykes on (*a*) AF demagnetization and (*b*) thermal demagnetization. Symbols as in Fig. 6.

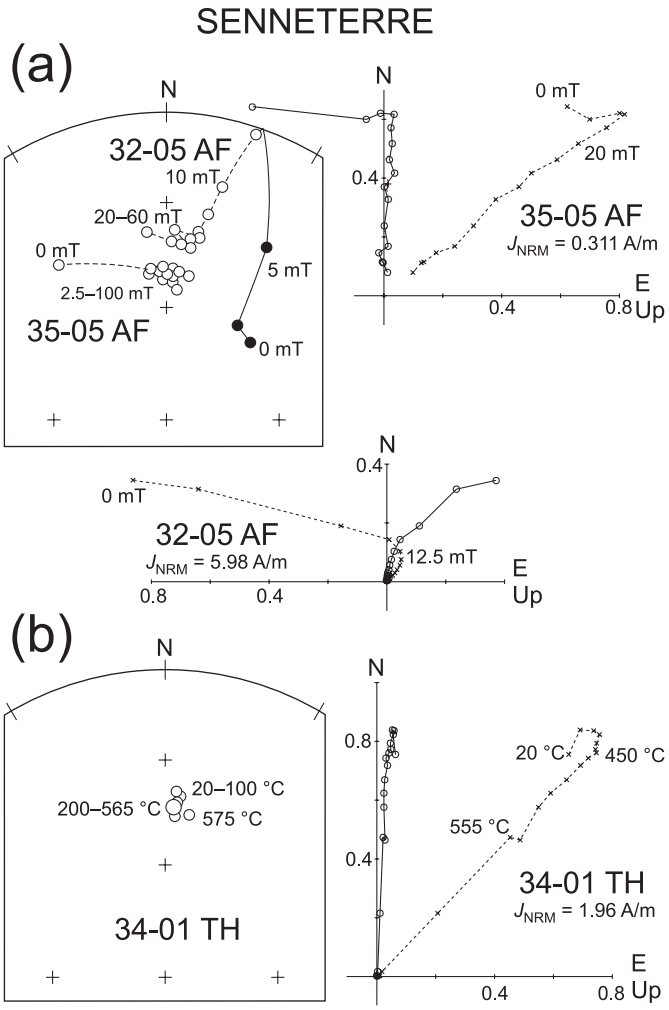
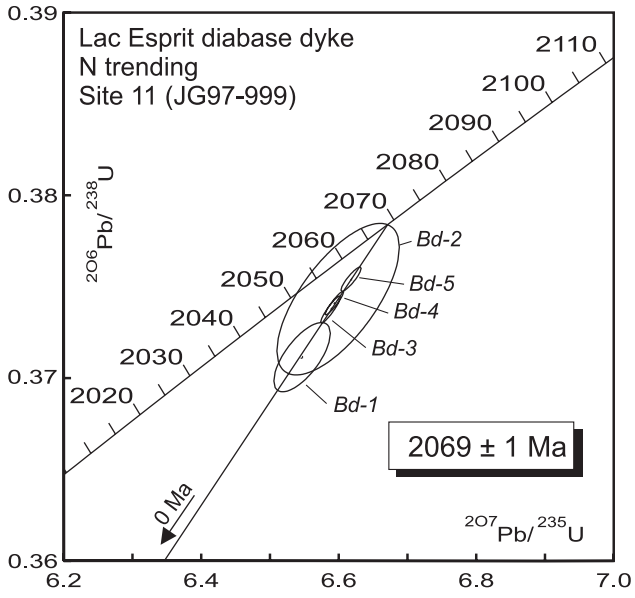


Fig. 10. Concordia diagram showing U–Pb geochronology of Lac Esprit dyke LE4 at site 11 (modified after Hamilton et al. 2001). Error ellipses are at the 2σ level of uncertainty.



study, the remanence of Lac Esprit dykes was likely acquired at the time of emplacement at 2069 Ma or shortly thereafter for the following reasons. Firstly, older Senneterre dykes that cross the area (Fig. 2) retain a primary remanence as discussed earlier. It is unlikely that Lac Esprit dykes could be completely overprinted without affecting the Senneterre dykes, especially since at most sites the two dyke sets have similar blocking temperature spectra.

Secondly, we interpret the Lac Esprit remanence as primary based on a positive secular variation test. In particular, mean paleomagnetic directions from sites along individual dykes (Table 3) usually form distinct clusters (Fig. 5*a*). This likely reflects secular variation of the Earth's field during dyke swarm emplacement, because individual dykes cool too rapidly to average out secular variation. Site mean directions would not group according to the dyke from which they were collected if the remanence was a later overprint, given the similar blocking temperature spectra of the different dykes.

As described earlier, the Mistassini swarm carries three superimposed remanence directions (Fig. 8*c*; Fahrig et al. 1986). Fahrig et al. concluded that the steep down component was likely primary, corresponding to an emplacement age >2000 Ma based on K–Ar dating of the swarm. However, more recently, a much older U–Pb age of 2510–2500 Ma has been reported for the Mistassini dykes (Heaman 2004). This suggests that the steep down component is not primary because it is widely divergent from the stable magnetization of 2505 Ma Ptarmigan dykes (Buchan et al. 1998; Fig. 1*b*). On the other hand, Fahrig et al.'s "northeast-down" magnetization has a VGP that is indistinguishable from that of the Ptarmigan magnetization (Fig. 14). Therefore, we propose that the down NE component is primary and the steeper component a later overprint. In the study area two sites (25 and 29) may carry primary magnetizations of opposite polarity. Further paleomagnetic study of Mistassini dykes is needed to test this interpretation.

Age constraints for Sakami Formation

Crosscutting relationships between the various dyke swarms of the sampling area and the Sakami Formation are important for constraining the age of the Sakami Formation and have been summarized by Goutier et al. (2001*a*). Northwest-trending Lac Esprit dyke LE7 cuts the Sakami Formation at site 22 (Fig. 4), demonstrating that the Sakami Formation predates intrusion of the Lac Esprit dykes at 2069 Ma. In addition, the ENE-trending dyke at site 39 that is interpreted to belong to Senneterre dyke SE2 (see earlier in the text) also cuts the Sakami Formation. This suggests that the Sakami Formation predates the 2216 Ma emplacement of the Senneterre swarm and is significantly older than the Richmond Gulf Group on the east coast of Hudson Bay, whose maximum age has been estimated at 2025 ± 25 Ma (Chandler and Parrish 1989).

Dyke MS2 (site 26) is overlain by the Sakami Formation. As noted previously, it has been interpreted to belong to the Mistassini swarm on the basis of geochemistry and trend. Thus the Sakami Formation postdates 2510–2500 Ma, the age of the Mistassini swarm, and is constrained to fall be-

© 2007 NRC Canada

Table 4. U–Pb analytical data for Lac Esprit north–south diabase dyke, site 11 (JG97-999; labelled JG97-EKN in Hamilton et al. (2001)).

Fraction			Concentrations				Atomic Ratios							
Description[a]	No. of grains	Wt.[b] (µg)	U[b] (ppm)	Pb*[b] (ppm)	Pb[c] (pg)	Th/U[d]	$\frac{^{206}Pb}{^{204}Pb}$	$\frac{^{208}Pb}{^{206}Pb}$	$\frac{^{206}Pb}{^{238}U}$	$\frac{^{207}Pb}{^{235}U}$	$\frac{^{207}Pb}{^{206}Pb}$	$\frac{^{207}Pb}{^{206}Pb}$ age (Ma)	Disc. (%)
Coarse-grained to pegmatitic gabbro (45 m wide dyke; GSC laboratory no. z5107; 53.380°N, 77.511°W)													
Bd-1 nm0°, sf	28	4	31	12	8	0.147	437	0.0423	0.3711%±0.26%	6.547%±0.31%	0.12793%±0.23%	2069.7±8.2	2.0
Bd-2 nm0°, sf	20	4	21	8	11	0.091	193	0.0260	0.3743%±0.55%	6.599%±0.67%	0.12786%±0.50%	2068.8±17.6	1.1
Bd-3 m0°, lb	4	3	116	43	2	0.100	3641	0.0288	0.3738%±0.10%	6.588%±0.11%	0.12784%±0.04%	2068.5±1.5	1.2
Bd-4 m0°, lb	40	16	68	25	7	0.078	3365	0.0225	0.3741%±0.09%	6.594%±0.10%	0.12783%±0.04%	2068.3±1.3	1.1
Bd-5 m0°, lb, abr 1.5 hrs	25	9	125	47	13	0.091	1775	0.0261	0.3754%±0.09%	6.618%±0.11%	0.12785%±0.05%	2068.7±1.8	0.8

Note: Errors are 1 std. error of mean in % except Pb^{207}/Pb^{206} age errors, which are 2 std. errors in Ma; *, Radiogenic Pb. Disc., percent discordance on discordia to origin.

[a]Mineral prefix: Bd, baddeleyite; abr, abraded. Shape: sf, short fragments; lb, long blades. Magnetic properties: nm0°, nonmagnetic at 0° side slope on Frantz™ magnetic separator at maximum current; m0°, magnetic at 0° side slope.

[b]Uncertainties in concentrations are estimated to be <10% for sample weights >10 µg, >10% for sample weights <10 µg. Uncertainties in sample weights estimated at ±1 µg.

[c]Total common Pb in analysis, corrected for spike and fractionation.

[d]Th/U from $^{208}Pb^*/^{206}Pb^*$ and $^{207}Pb^*/^{206}Pb^*$ ages.

tween 2500 and 2216 Ma, similar to the 2450–2217 Ma age bracket for the Huronian Supergroup of the Southern Province.

Dyke swarm distributions

The extent of the Lac Esprit dyke swarm as currently mapped is shown in Fig. 2. Beyond the sampling area its distribution is unclear because there is a lack of detailed mapping to the north and because trend alone is not a conclusive distinguishing characteristic. To the east and southeast, northwest-trending Lac Esprit dykes are difficult to separate from Mistassini dykes. Similarly, north-trending Lac Esprit dykes may be confused with Matachewan or Lac Shpogan dykes.

The north to northwest range in trend of Lac Esprit dykes can be explained in several ways. These include emplacement over some time interval during which the regional stress field changed, the influence of basement structures on orientations (e.g., Jourdan et al. 2006) or a conjugate dyke set.

The Matachewan swarm has previously been mapped over a large area south and west of James Bay. This study yields the first demonstration that the swarm also extends into the area east of James Bay (sites 09 and 15), as indicated by the similarity between the VGP for sites 09 and 15 and well-defined Matachewan paleopoles (Fig. 14).

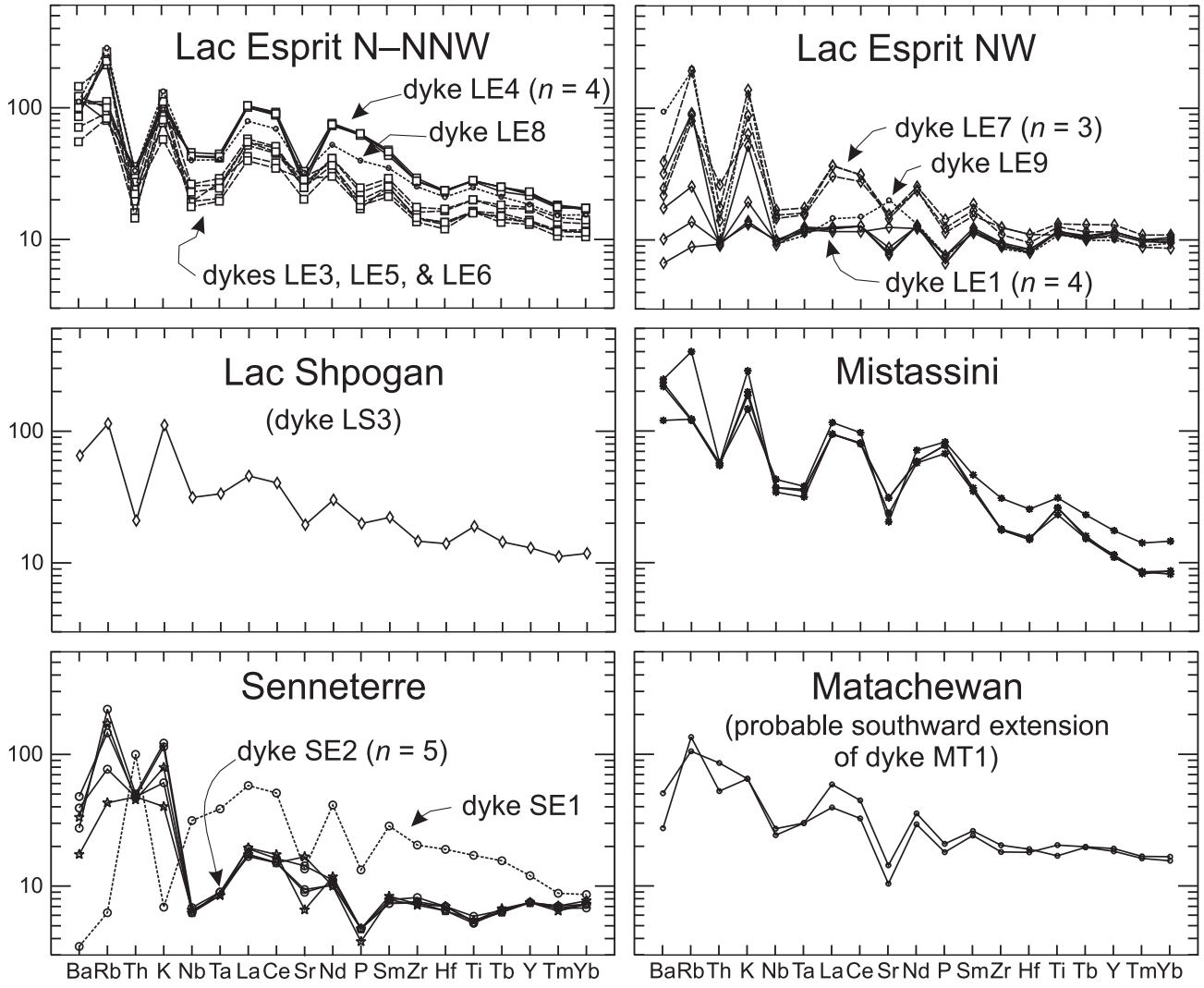
Buchan et al. (1998) proposed that Senneterre, Maguire, and Klotz dykes of Fig. 1(*b*) define a giant radiating dyke swarm (the Ungava swarm) with a focus off the eastern margin of the Superior Province near Ungava Bay. They also suggested that Nipissing sills of the Southern Province (Fig. 15) were fed by coeval Senneterre dykes from a plume centre beneath the swarm's focal point. Senneterre dykes of the study area have northeast trends that fall between the NNE trends of Senneterre dykes to the south and east trends of Maguire dykes to the north (Fig. 15). They help to fill out and better define the overall radiating pattern of the giant Ungava swarm, which fans across the eastern Superior Province from the Grenville Front in the south to the Cape Smith Belt in the north.

Missing portions of the giant Ungava swarm may currently be located on cratonic blocks that rifted away from the northeastern and southeastern portion of the Superior Province in the Paleoproterozoic. As yet these missing portions of the swarm have not been identified. Roughly coeval magmatism is found on several Archean cratonic blocks around the world including MacKay dykes in the Slave Province of the Canadian Shield (LeCheminant and van Breemen 1994), Ongeluk volcanic rocks in the Kaapvaal craton of southern Africa (Cornell et al. 1996), sills in the Karelian Province of Finland (Vuollo and Huhma 2005) and sills in the Hamersley Province of western Australia (Müller et al. 2005). However, locating the cratonic blocks relative to the Superior Province will require paleopoles for these magmatic units that are demonstrated to be primary.

The overall distribution of Lac Shpogan dykes is unknown. The Lac Shpogan VGP is distinct from other Superior Province paleopoles. It falls ~25° south of the paleopole for Minto dykes (Fig. 14). Because secular variation has not been averaged out in the Lac Shpogan pole, a correlation with the Minto dykes cannot be ruled out. Mesoproterozoic paleopoles for the 1238 Ma Sudbury dyke swarm of the southern Superior and Southern provinces and the 1267 Ma

© 2007 NRC Canada

Fig. 11. Spidergrams showing data from dykes of this study. Values are chondrite-normalized as in Thompson et al. (1983).



Mackenzie swarm of the northern Canadian Shield (poles not shown in Fig. 14) lie ~25° and 15° southeast of the Lac Shpogan pole respectively. However, they are consistently of opposite polarity to the Lac Shpogan pole suggesting that a correlation with either of these swarms is unlikely. Finally, the Neoproterozoic paleopole for the 723 Ma Franklin dyke swarm of the northern Canadian Shield (pole not shown in Fig. 14) falls ~15° southwest of the Lac Shpogan pole. Franklin dykes are of dual polarity and have been identified in the extreme northeastern portion of the Superior Province where they trend northwest (e.g., Buchan and Ernst 2004). However, Lac Shpogan and Franklin dykes have somewhat dissimilar geochemical characteristics (Figs. 12, 13). Further work is required to establish the age of Lac Shpogan dykes and better define their paleomagnetic direction.

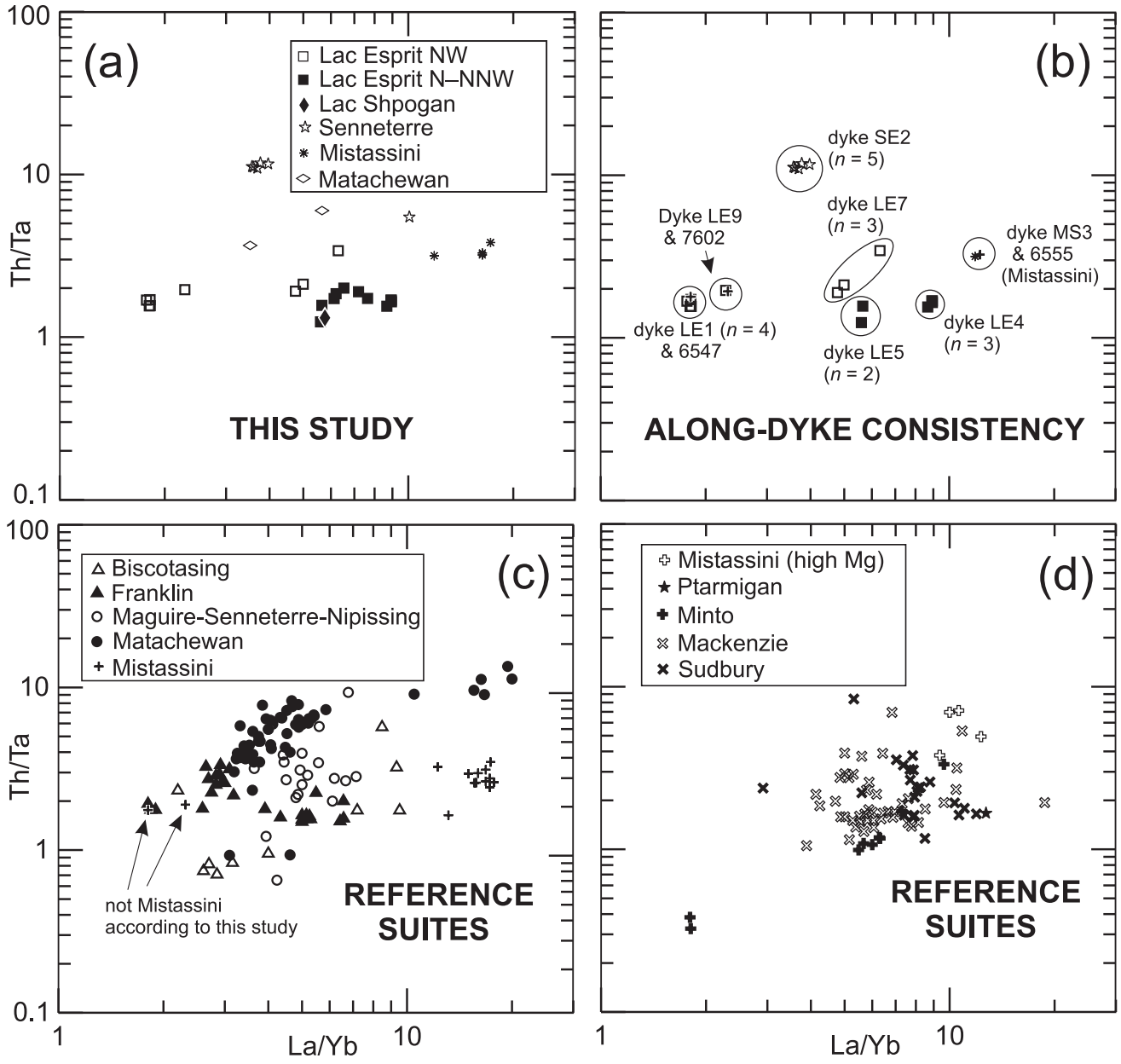
Vertical axis rotations within Superior Province

Paleoproterozoic paleopoles for the Superior Province define a progression from southeast to northwest between 2220 and 2070 Ma (Fig. 14), with the Lac Esprit paleopole of this study falling close to, but distinct from, the similar-aged Fort Frances paleopole of Halls (1986). As illustrated in Fig. 16, where the paleomagnetic directions for the two swarms are corrected to a common reference locality, the difference between the Lac Esprit and Fort Frances paleopoles reflects a difference in magnetic declination for the two dyke swarms of 23° ± 12° (calculated using the method of Beck 1980) and a difference of ~8.5° in inclination. The difference in declination is unlikely to reflect secular variation because there is a sufficient number of dykes in each swarm to largely average out secular variation. It could be interpreted in terms of very rapid apparent polar wander and indicate a ~23° clockwise rotation of the Superior Province between the emplacement of Fort Frances dykes at 2076^{+5}_{-4} Ma and Lac Esprit dykes at 2069 ± 1. However, given the similar emplacement ages, a more likely interpretation is that vertical axis rotation has occurred between the two geographical areas in the eastern and western Superior Province since dyke emplacement, with the block containing the Fort Frances dykes having rotated counterclockwise ~23° relative to that hosting the Lac Esprit dykes. The latter interpretation is discussed in more detail in a subsequent section.

Two giant Paleoproterozoic dyke swarms, the 2473–2446 Ma Matachewan and 2170–2167 Ma Biscotasing swarms, are

© 2007 NRC Canada

Fig. 12. Th/Ta vs. La/Yb diagram (Condie 1997). (*a*, *b*) This study. (*c*, *d*) Reference suites. High-Mg Mistassini dykes of the Lac Mistassini area are subparallel to 2510–2500 Ma tholeiitic Mistassini dykes (Fahrig et al. 1986), but have not been dated and hence may be of a different age. Dykes 6547 and 7602 south of the sampling area (see Fig. 2), which were interpreted to belong to the Mistassini swarm by Fahrig et al. (1986), are reinterpreted in this study to belong to the Lac Esprit swarm.

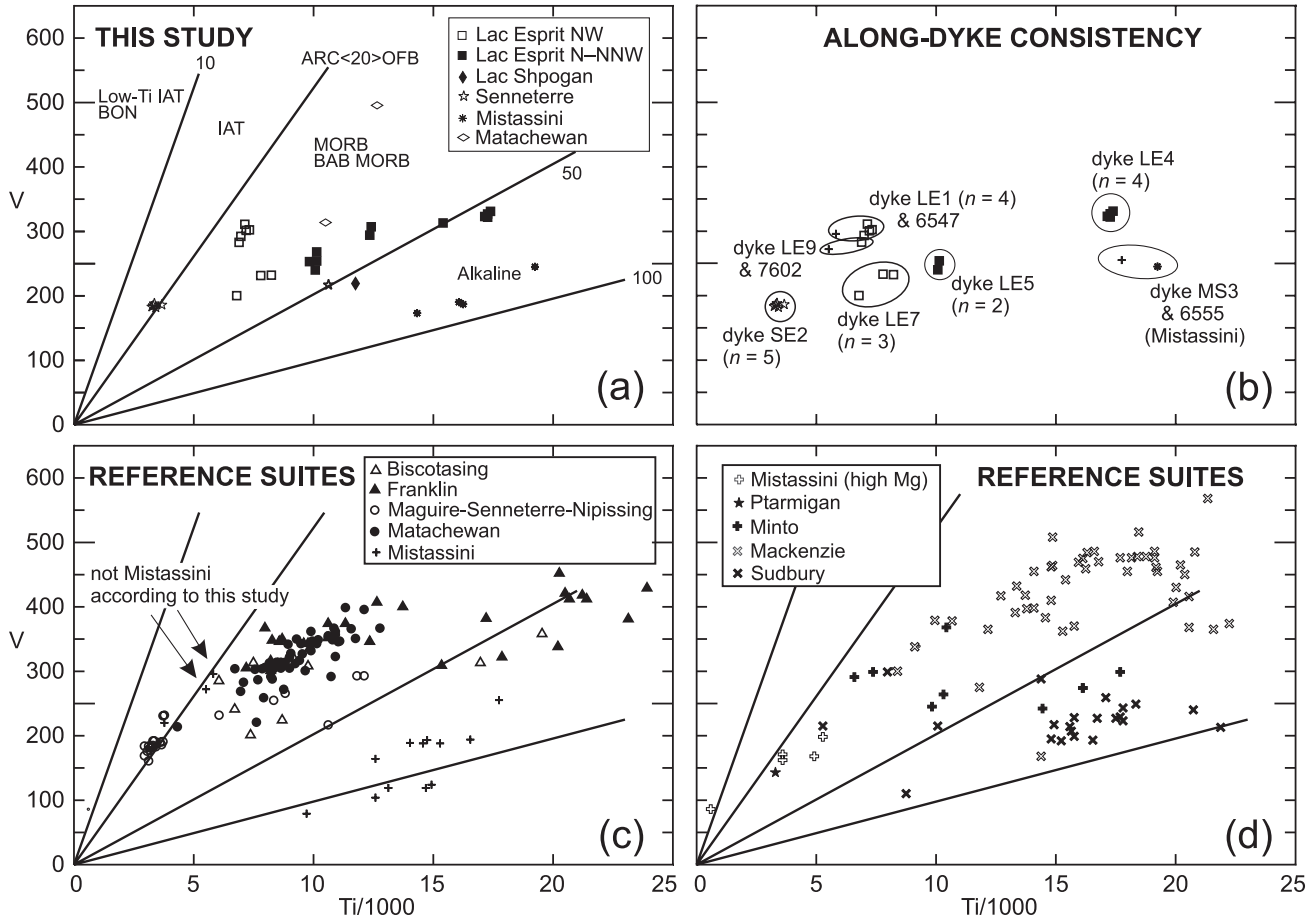


bisected by the Kapuskasing Structural Zone (KSZ), a prominent zone of crustal uplift separating the western and eastern portions of the Superior Province (Fig. 1; Percival and West 1994). Within a region that extends ~400 km northwest of the KSZ, differential rotation of several blocks is evident in both Matachewan paleomagnetic declinations and Matachewan dyke orientations (Bates and Halls 1991; West and Ernst 1991). West and Ernst re-aligned Matachewan dyke aeromagnetic anomalies on either side of the KSZ on the assumption that the primary geometry of the swarm is that of a giant fan. Their interpretation indicated ~60–80 km of right lateral displacement along the KSZ since Matachewan dyke emplacement. Bates and Halls (1991) and Halls et al. (2005) confirmed the rotation of blocks northwest of the KSZ by demonstrating that the magnetic declinations of Matachewan dykes in several blocks have rotated relative to one another.

Halls and Davis (2004) compared Matachewan magnetic declinations in the three areas from the Bates and Halls (1991) study that are located farthest to the north and west (Hearst, Hornepayne, and Ogoki Lake areas of Fig. 1*a*) with those of Matachewan dykes east of the KSZ. They concluded that the Hearst – Hornepayne – Ogoki Lake area has rotated counterclockwise 12° ± 9° (Fig. 16) around a vertical axis relative to the area east of the KSZ since the time of dyke emplacement at 2473–2446 Ma. Subsequently, analysis

© 2007 NRC Canada

Fig. 13. V vs. Ti diagram (Shervais 1982). (*a*, *b*) This study. (*c*, *d*) Reference suites. V and Ti are in ppm.



of Matachewan dyke data in the Lake Nipigon – Ogoki Lake area (Fig. 1*a*) has yielded magnetic declinations (Halls et al. 2005) that are similar to those of the Hornepayne area, confirming the earlier interpretation.

Halls and Davis (2004) also compared paleomagnetic data from Biscotasing dykes west of the KSZ with earlier data of Buchan et al. (1993) collected east of the KSZ. They estimated a counterclockwise rotation of 16° ± 21° for the area west of the KSZ relative to the area east of the zone (Fig. 16) since dyke emplacement at ca. 2170 Ma. A post-2170 Ma age for the deformation across the KSZ is in agreement with the earlier interpretation of Buchan and Ernst (1994) based on the observation that the most prominent Onaping fault which offsets the Biscotasing dykes is in turn offset by movement along the KSZ. The large uncertainty (±21°) for the Biscotasing rotation is due in part to the steep remanence directions of the Biscotasing dykes (steep directions are less sensitive to vertical axis rotations). Halls and Davis also noted that the mean trend of Biscotasing dykes west of the KSZ is rotated about 21° counterclockwise from the mean trend east of the KSZ, perhaps as a result of the proposed block rotation.

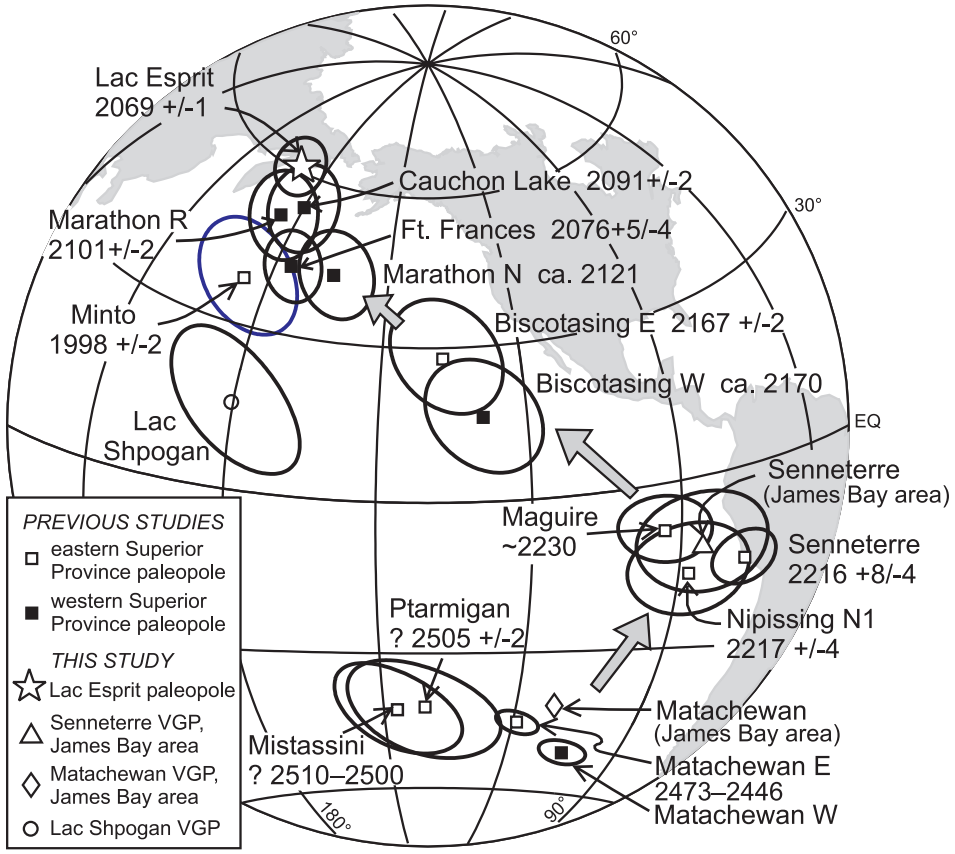
The 23° ± 12° difference in paleomagnetic declination for Lac Esprit and Fort Frances dykes (Fig. 16; Table 2) in this study may also reflect vertical axis rotation across the KSZ after dyke emplacement at ~2070 Ma. The Lac Esprit and Fort Frances areas are separated by ~1500 km, and are far from those studied closer to the KSZ by Halls and Davis (2004). Nevertheless, the sense of rotation is the same as Halls and Davis proposed based on Matachewan and Biscotasing dykes, with the Fort Frances area of the western Superior Province rotated counterclockwise to the Lac Esprit area of the eastern Superior Province. This suggests that broad-scale differential rotation may have affected not only the region within 400 km of the KSZ but the entire western and eastern portions of the Superior Province (Fig. 17).

Unfortunately, no single early Paleoproterozoic dyke swarm (or set of similar-aged swarms) has yet been identified that extends over the western Superior Province. The paleomagnetic signature and geometry of such a swarm would allow an assessment of block rotations over the whole area. Nevertheless, some additional information may be obtained by comparing the paleomagnetic data from 2091 ± 2 Ma (Halls and Heaman 2000) Cauchon Lake dykes near the Trans-Hudson Orogen and 2101 ± 2 Ma (Hamilton et al. 2002) “reversely” magnetized Marathon dykes north of Lake Superior (Fig. 1*b*), whose ages differ by only about 10 million years. Paleopoles for these two swarms are similar (Fig. 14), indicating that there has been little relative rotation between these widespread areas of the western Superior Province after 2091 Ma, and supporting the proposal that much of the western Superior Province rotated as a block relative to the eastern Superior Province.

As noted earlier, in the western Superior Province within

© 2007 NRC Canada

Fig. 14. Paleopoles dated at ~2.51–2.00 Ga, ovals of 95% confidence and U–Pb ages (in Ma) from Superior and Southern provinces. Pole locations are well constrained except for Ptarmigan, Mistassini, Senneterre (James Bay area), Matachewan (James Bay area) and Lac Shpogan virtual geomagnetic poles (VGPs), which are based on a very limited number of dykes. Question marks before Ptarmigan and Mistassini ages reflect lack of tests demonstrating that paleopoles are primary. Open (closed) symbols indicate paleopoles from units east (west) of the Kapuskasing Structural Zone. Large arrows show the trend of paleopole ages, not an apparent polar wander path. See text for reinterpretation of the Mistassini paleopole (based on the NE down magnetization component) as possibly primary. References for paleopoles and ages are: Mistassini, Fahrig et al. (1986) and Heaman (2004); Matachewan, Bates and Halls (1990) and Heaman (1997); Nipissing N1 sills, Buchan et al. (1998, fig. 11) and Noble and Lightfoot (1992); Senneterre, Buchan et al. (1993, 1996); Ptarmigan, Maguire, and Minto, Buchan et al. (1998); Biscotasing, Buchan et al. (1993) and Halls and Davis (2004); Marathon N and Marathon R, Buchan et al. (1996) and Hamilton et al. (2002); Fort Frances, Halls (1986) and Buchan et al. (1996); Cauchon Lake, Halls and Heaman (2000); Lac Esprit, Lac Shpogan, Matachewan (James Bay area), and Senneterre (James Bay area), this study.



400 km of the KSZ the paleomagnetic data and geometry of the Matachewan dyke swarm indicate that more complex rotations have occurred (Bates and Halls 1991). Whether this distortion of the swarm is actually due to rotation of rigid blocks or to more distributed distortion (Riller et al. 1999, p. 65) is unclear.

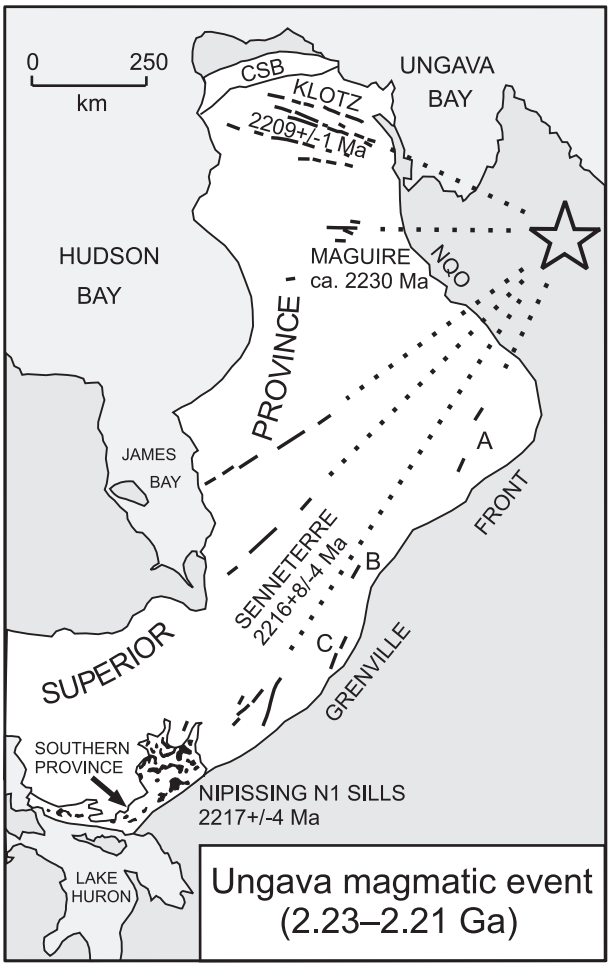
The integrity of the eastern Superior Province since the early Paleoproterozoic is easier to assess than that of the western Superior Province, because the Ungava radiating dyke swarm has now been identified over much of the eastern portion of the province (Fig. 15). The consistent fanning geometry (Fig. 15) and paleopoles (Fig. 14) from the swarm demonstrate that no substantial block rotations have occurred over most of the eastern Superior Province since dyke emplacement at ca. 2220–2210 Ma.

More limited data on the integrity of the eastern Superior Province come from older 2505 Ma Ptarmigan and ~2510–2500 Ma Mistassini swarms (Fig. 1*a*). Paleopoles that we suggest may be primary for these two units (Fig. 14) are indistinguishable, indicating no relative rotation of the Ptarmigan and Mistassini areas since 2500 Ma. More work on dykes of this age, however, is required to confirm this suggestion.

The timing of relative rotation of the eastern and western Superior Province is constrained in this study to have occurred after 2069 Ma, the age of Lac Esprit dyke emplacement. The complex deformation that is recorded in the western Superior Province within 400 km of the KSZ is likely coeval with this overall rotation event. In any case, it must have occurred after 2170 Ma, the age of Biscotasing dykes that were deformed along with the older Matachewan swarm in this region. Major uplift within the KSZ is thought to be associated with the complex rotations that occurred to the northwest (Halls and Davis 2004). Several authors have discussed data from within the KSZ that could help constrain the timing of major uplift, although, as yet, no consensus has emerged. Maximum uplift ages vary from ~2043 Ma, the Ar–Ar age of Kapuskasing dykes (Hanes et al. 1994) that are thought to

© 2007 NRC Canada

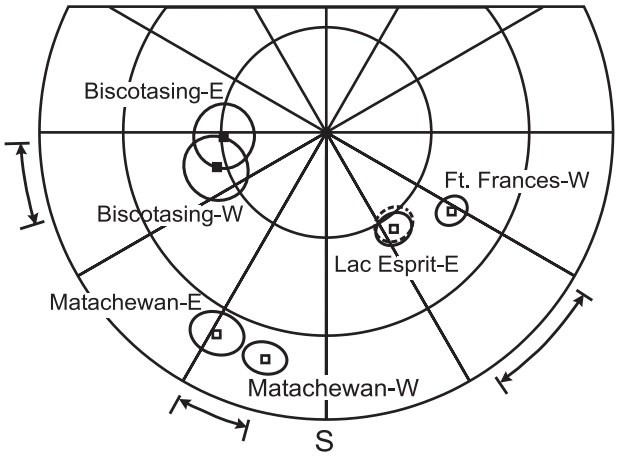
Fig. 15. Ungava magmatic event of eastern Superior and Southern provinces. Dotted lines represent trends of dyke sets radiating from a focal point (star) that may locate a mantle plume centre. Dyke locations and trends are mainly after Buchan et al. (1993, 1998) and this study. Dykes labelled A are after Ernst et al. (1998). The dyke labelled B refers to site 7606 in Fig. 2 of Fahrig et al. (1986). Dykes labelled C are in the Lacs Obatogamau area. Dykes B and C are interpreted to be Senneterre dykes based on paleomagnetic data in unpublished files of W.F. Fahrig that are held at the Geological Survey of Canada. Nipissing sills intruded Huronian Supergroup sedimentary rocks and may have been fed laterally via Senneterre dykes. CSB and NQO are Cape Smith Belt and New Quebec Orogen, respectively.



be deeply exhumed based on paleomagnetism (Symons et al. 1994) and amphibole geobarometry (Percival et al. 1994), to 1.93 Ma, the youngest Rb–Sr biotite ages obtained from Archean granitoid rocks (Percival and Peterman 1994). See a more detailed discussion in the recent paper by Halls and Davis (2004). The minimum age of the uplift (and rotational) event is even more poorly constrained. As yet, no geological units of identical age on both sides of the KSZ have been identified that do not show significant relative rotation. In addition, no single geological unit has been found that clearly crosses the KSZ itself without distortion.

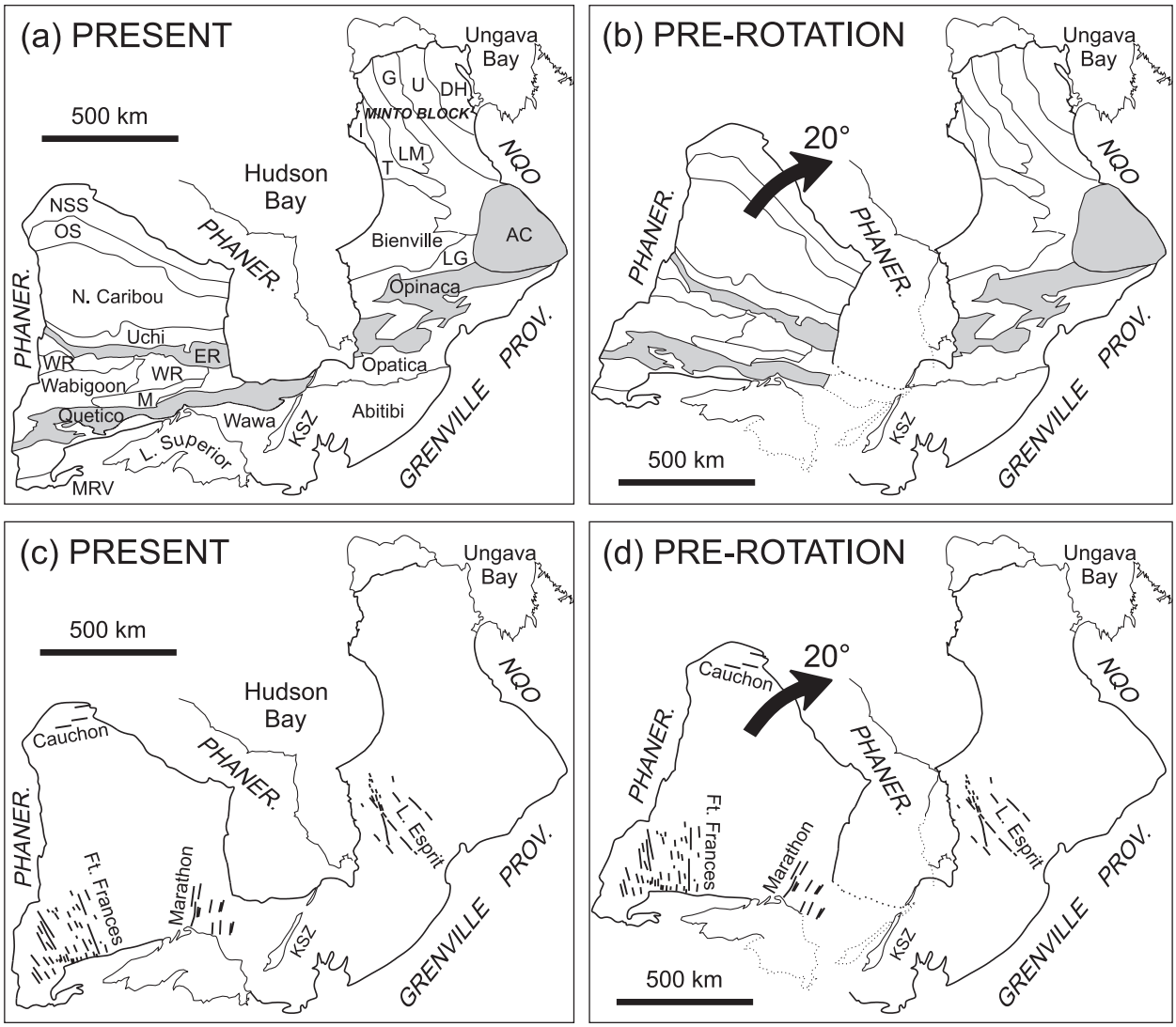
Several possible causes for relative rotation of the eastern and western Superior Province have been proposed, each dependent on the timing of rotation. Bates and Halls (1991) concluded that deformation across the KSZ was probably related to the ~1950 Ma (now estimated at ~1850–1800 Ma) Trans-Hudson Orogeny along the northwestern margin of the Superior craton. Alternatively, the deformation could be related to the ~1.89–1.83 Ga Penokean Orogeny on the southern margin of the Superior craton, perhaps the result of indentation of the southern Superior by the Wisconsin magmatic terrane (e.g., Riller et al. 1999). Most recently, Halls and Davis (2004) have proposed that relative rotation of the eastern and western Superior Province may be related to ~2000 Ma continental rifting beneath Hudson Bay. A major magmatic event(s) of this age occurs in the eastern Hudson Bay region. Based on geochronological and geochemical evidence (Schwarz and Fujiwara 1981; Parrish 1989; Legault et al. 1994; Buchan et al. 1998), it may include the Minto and Inukjuak dyke swarms east of Hudson Bay, Eskimo volcanics of the Belcher Islands, Persillon volcanics of Richmond Gulf, Povungnituk volcanics of the western Cape Smith Belt, and the Purtuniq ophiolite of the Cape Smith Belt. Further evidence for or against a rift beneath Hudson Bay and better constraints on timing of the relative rotation of the two portions of the Superior Province would help distinguish among these models. In particular, a comparison of paleopoles for ~1880 Ma magmatism that is known from both the western and eastern Superior Province would clarify whether rotation occurred before or after 1880 Ma and, hence, would differentiate between an ~2000 Ma rift model and one related to the ~1850–1800 Ma Trans-Hudson Orogeny.

Fig. 16. Mean paleomagnetic directions and circles of 95% confidence plotted on an equal area net for dyke swarms east (E) and west (W) of the Kapuskasing Structural Zone (KSZ) in the Superior Province. Closed (open) symbols indicating down (up) directions. Lac Esprit and Fort Frances data are corrected to a common reference locality at 53.5°N, 77.0°W. Solid and dashed circles of 95% confidence for the Lac Esprit mean direction refer to all dykes and only north- to NNW-trending dykes, respectively. Biscotasing and Matachewan dyke data are from Buchan et al. (1993), Halls and Davis (2004) and Halls et al. (2005) and are corrected to a common reference locality at 48°N, 79°W. Matachewan data west of the KSZ are from the Lake Nipigon (LN) – Ogoki Lake (OL) area of Fig. 1*a*. The difference in declination for mean directions from similar-aged dykes on either side of the KSZ is indicated by the arrows outside the equal area net.



© 2007 NRC Canada

Fig. 17. Superior Province (*a*, *c*) in present configuration and (*b*, *d*) after correction for ~20° relative rotation of eastern and western portions of the province based on paleomagnetic data and ~70 km of right lateral displacement along the Kapuskasing Structural Zone (KSZ). NQO, New Quebec Orogen; Phaner., Phanerozoic cover. Terranes are modified after Percival et al. (1992, in preparation[3]) and Leclair et al. (2004) and are labelled as follows: AC, Ashuanipi Complex; DH, Douglas Harbour; ER, English River; G, Goudalie; I, Inukjuak; LG, La Grande; LM, Lac Minto; M, Marmion; NSS, Northern Superior superterrane; OS, Oxford-Stull; T, Tikkerutuk; U, Utsalik; WR, Winnipeg River. Metasedimentary terranes are shaded. In (*b*) and (*d*), geological and geographical elements are dotted in the region where rather complicated block rotations have been reported west of the KSZ (Bates and Halls 1991). In (*c*) and (*d*) the distribution of the Fort Frances and Lac Esprit swarms, whose paleomagnetic data suggest relative rotation of the eastern and western Superior Province, are illustrated. Also shown are 2101 Ma "reversely" magnetized Marathon and 2091 Ma Cauchon dykes, whose paleomagnetic data indicate little or no relative rotation within the western Superior Province.



Conclusions

The Lac Esprit dyke swarm of the eastern Superior Province yields a U–Pb emplacement age of 2069 ± 1 Ma, similar to the 2076^{+5}_{-4} Ma age of the Fort Frances dyke swarm ~1500 km away in the western Superior Province. Primary paleomagnetic declinations from the two swarms (when reduced to a common reference locality) differ by 23° ± 12°, likely reflecting relative rotation of the eastern and western portions of the province after 2069 Ma. This supports and augments earlier paleomagnetic studies that reported post-2170 Ma relative rotations in a geographically more restricted area within 400 km of the KSZ separating the eastern and western Superior Province.

Two prominent Senneterre dykes identified in the study area help to define a giant swarm that spans the eastern Superior Province from the Grenville Front in the south to the Cape Smith Belt in the north. The simple radiating geometry of the swarm and consistent paleomagnetic directions indicate that no significant block rotations have occurred

[3] J.A. Percival, M. Sanborn-Barrie, T. Skulski, G.M. Stott, and H. Helmstaedt. Tectonic evolution of the Western Superior Province from NATMAP and Lithoprobe studies. In preparation.

© 2007 NRC Canada

within the corresponding area of the eastern Superior Province since dyke emplacement at ~2220–2210 Ma.

Crosscutting relationships indicate that the Sakami Formation is younger than the ~2510–2500 Ma Mistassini dykes, but older than the 2069 Ma Lac Esprit and 2216 Ma Senneterre dykes.

Acknowledgements

We thank Michel Houlé and Patrick Mercier-Langevin for assistance in the field and John Morgan for help in the paleomagnetic laboratory. John Percival, reviewers Steve Harlan and Sergei Pisarevsky, and Associate Editor Bill Davis provided useful comments that have helped to improve the manuscript. Figure 10 and Table 4 are reproduced with permission of the Minister of Public Works and Government Services Canada, 2006, and courtesy of Natural Resources Canada, Geological Survey of Canada.

References

Bailey, R.C., and Halls, H.C. 1984. Estimates of confidence on paleomagnetic directions derived from mixed remagnetization circle and direct observational data. Journal of Geophysics, **54**: 174–182.

Bates, M.P., and Halls, H.C. 1990. Regional variation in paleomagnetic polarity of the Matachewan dyke swarm related to the Kapuskasing Structural Zone, Ontario. Canadian Journal of Earth Sciences, **27**: 200–211.

Bates, M.P., and Halls, H.C. 1991. Broad-scale Proterozoic deformation of the central Superior Province revealed by paleomagnetism of the 2.45 Ga Matachewan dike swarm. Canadian Journal of Earth Science, **28**: 1780–1796.

Beck, M.E. 1980. Paleomagnetic record of plate-margin tectonic processes along the western edge of North America. Journal of Geophysical Research, **85**: 7115–7131.

Buchan, K.L. 1991. Baked contact test demonstrates primary nature of dominant (N1) magnetisation of Nipissing intrusions in Southern Province, Canadian Shield. Earth and Planetary Science Letters, **105**: 492–499.

Buchan, K.L., and Ernst, R.E. 1994. Onaping fault system: age constraints on deformation of the Kapuskasing structural zone and units underlying the Sudbury Structure. Canadian Journal of Earth Sciences, **31**: 1197–1205.

Buchan, K.L., and Ernst, R.E. 2004. Diabase dyke swarms and related units in Canada and adjacent regions. Geological Survey of Canada, Map 2022A, scale 1 : 5 000 000.

Buchan, K.L., Mortensen, J.K., and Card, K.D. 1993. Northeast-trending Early Proterozoic dykes of southern Superior Province: multiple episodes of emplacement recognized from integrated paleomagnetism and U–Pb geochronology. Canadian Journal of Earth Sciences, **30**: 1286–1296.

Buchan, K.L., Halls, H.C., and Mortensen, J.K. 1996. Paleomagnetism, U–Pb geochronology, and geochemistry of Marathon dykes, Superior Province, and comparison with the Fort Frances swarm. Canadian Journal of Earth Sciences, **33**: 1583–1595.

Buchan, K.L., Mortensen, J.K., Card, K.D., and Percival, J.A. 1998. Paleomagnetism and U–Pb geochronology of diabase dyke swarms of Minto Block, Superior Province, Quebec, Canada. Canadian Journal of Earth Sciences, **35**: 1054–1069.

Chandler, F.W., and Parrish, R.R. 1989. Age of the Richmond Gulf Group and implications for rifting in the Trans-Hudson Orogen, Canada. Precambrian Research, **44**: 277–288.

Condie, K.C. 1997. Sources of Proterozoic mafic dyke swarms: constraints from Th/Ta and La/Yb ratios. Precambrian Research, **81**: 3–14.

Condie, K.C. 2003. Incompatible element ratios in oceanic basalts and komatiites: Tracking deep mantle sources and continental growth rates with time. Geochemistry, Geophysics, Geosystems. **4**(1): 1005, doi: 10.1029/2002GC000333.

Cornell, D.H., Schütte, S.S., and Eglington, B.L. 1996. The Ongeluk basaltic andesite formation in Griqualand West, South Africa: submarine alteration in a 2222 Ma Proterozoic sea. Precambrian Research, **79**: 101–123.

Eade, K.E. 1966. Fort George and Kaniapiskau River (west half) map-areas, New Quebec. Geological Survey of Canada, Memoir 339.

Ernst, R.E., Buchan, K.L., Goutier, J., Leclair, A., and Lamothe, D. 1998. Reconnaissance paleomagnetic study of diabase dykes of the James Bay and Ashuanipi regions of Quebec. Geological Association of Canada, Abstract Vol., p. 53.

Fahrig, W.F., Gaucher, E.H., and Larochelle, A. 1965. Paleomagnetism of diabase dykes of the Canadian Shield. Canadian Journal of Earth Sciences, **2**: 278–298.

Fahrig, W.F., Christie, K.W., Chown, E.H., Janes, D., and Machado, N. 1986. The tectonic significance of some basic dyke swarms in the Canadian Superior Province with special reference to the geochemistry and paleomagnetism of the Mistassini swarm, Quebec, Canada. Canadian Journal of Sciences, **23**: 238–253.

Fisher, R.A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, **A217**: 295–305.

Goutier, J., Doucet, P., Dion, C., Beausoleil, C., David, J., Parent, M., and Dion, D.-J. 1998*a*. Géologie de la région du lac Kowskatehkakmow (33F/06). Ministère des Ressources naturelles, Québec, RG 98-16.

Goutier, J., Doucet, P., Dion, C., Beausoleil, C., and Dion, D.-J. 1998*b*. Géologie de la région du lac Esprit (33F/05). Ministère des Ressources naturelles, Québec, RG 98-09.

Goutier, J., Dion, C., David, J., and Dion, D.-J. 1999*a*. Géologie de la région de la passe Shimusuminu et du lac Vion (33F/11 et 33F/12). Ministère des Ressources naturelles, Québec, RG 98-17.

Goutier, J., Dion, C., Lafrance, I., David, J., Parent, M., and Dion, D.-J. 1999*b*. Géologie de la région des lacs Langelier et Threefold (SNRC 33F/03 et 33F/04). Ministère des Ressources naturelles, Québec, RG 98-18.

Goutier, J., Dion, C., Ouellet, M.-C., David, J., and Parent, M. 2000. Géologie de la région des lacs Guillaumat et Sakami (SNRC 33F/02 et 33F/07). Ministère des Ressources naturelles, Québec, RG 99-15.

Goutier, J., Dion, C., and Ouellet, M.-C. 2001*a*. Géologie de la région de la colline Bezier (33G/12) et du lac de la Montagne du Pin (33G/13). Ministère des Ressources naturelles, Québec, RG 2001-13.

Goutier, J., Dion, C., Ouellet, M.-C., Mercier-Langevin, P., and Davis, D.W. 2001*b*. Géologie de la colline Masson, de la passe Awapakamich, de la baie Carbillet, et de la passe Pikwahipanan (SNRC 33F/09, 33F/10, 33F/15, et 33F/16). Ministère des Ressources naturelles, Québec, RG 2001-10.

Goutier, J., Dion, C., Ouellet, M.-C., Davis, D.W., David, J., and Parent, M. 2002. Géologie de la région du lac Guyer (33G/05, 33G/06, et 33G/11). Ministère des Ressources naturelles, Québec, RG 2001-15.

Halls, H.C. 1986. Paleomagnetism, structure, and longitudinal correlation of Middle Precambrian dykes from northwestern Ontario and Minnesota. Canadian Journal of Earth Sciences, **23**: 142–157.

Halls, H.C., and Davis, D. 2004. Paleomagnetic and U–Pb geochronology of the 2.17 Ga Biscotasing dyke swarm, Ontario,

© 2007 NRC Canada

Canada: evidence for vertical-axis crustal rotation across the Kapuskasing Zone. Canadian Journal of Earth Sciences, **41**: 255–269.

Halls, H.C., and Heaman, L.M. 2000. The paleomagnetic significance of new U–Pb age data from the Molson dyke swarm, Cauchon Lake area. Canadian Journal of Earth Sciences, **37**: 87–103.

Halls, H.C., and Zhang, B. 2003. Crustal uplift in the southern Superior Province, Canada, revealed by paleomagnetism. Tectonophysics, **362**: 123–136.

Halls, H.C., Stott, G.M., and Davis, D.W. 2005. Paleomagnetism, geochronology, and geochemistry of several Proterozoic mafic dike swarms in northwestern Ontario. Ontario Geological Survey, Open File Report 6171.

Hamilton, M.A., Goutier, J., and Matthews, W. 2001. U–Pb baddeleyite age for the Paleoproterozoic Lac Esprit dyke swarm, James Bay region, Quebec. Radiogenic Age and Isotopic Studies, Report 14, Geological Survey of Canada, Current Research 2001-F5.

Hamilton, M.A., Davis, D.W., Buchan, K.L., and Halls, H.C. 2002. Precise U–Pb dating of reversely magnetized Marathon diabase dykes and implications for emplacement of giant dyke swarms along the southern margin of the Superior Province, Ontario. Radiogenic Age and Isotopic Studies, Report 15, Geological Survey of Canada, Current Research 2002-F6.

Hanes, J.A., Archibald, D.A., Queen, M., and Farrar, E. 1994. Constraints from $^{40}Ar/^{39}Ar$ geochronology on the tectonothermal history of the Kapuskasing uplift in the Canadian Superior Province. Canadian Journal of Earth Sciences, **31**: 1146–1171.

Heaman, L.M. 1997. Global mafic magmatism at 2.45 Ga: remnants of an ancient large igneous province? Geology, **25**: 299–302.

Heaman, L.M. 2004. 2.5–2.4 Ga global magmatism: remnants of supercontinents or products of superplumes. Geological Society of America Abstracts, **36**(6): 255 (Paper 105-5, Denver Conference, November 2004).

Hofmann, A.W. 1997. Mantle geochemistry: the message from oceanic volcanism. Nature, **385**: 219–229.

Jourdan, F., Féraud, G., Bertrand, H., Watkeys, M.K., Kampunzu, A.B., and Le Gall, B. 2006. Basement control on dyke distribution in Large Igneous Provinces: case study of the Karoo triple junction. Earth and Planetary Science Letters, **241**: 307–322.

Kirschvink, J.K. 1980. The least squares line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

LeCheminant, A.N., and van Breemen, O. 1994. U–Pb ages of Proterozoic dyke swarms, Lac de Gras area, N.W.T.: evidence for progressive break-up of an Archean supercontinent. Geological Association of Canada / Mineralogical Association of Canada, Program with Abstracts, p. 62.

Leclair, A., Berclaz, A., and David, J. 2004. The northeastern Superior Province in Quebec's Far North: A regional synthesis. The Celebratory Conference,Toronto, Ontario, Oct. 12–15, 2004, Lithoprobe Report 86, Program and Abstracts, 4 p.

Legault, F., Francis, D., Hynes, A., and Budkewitsch, P. 1994. Proterozoic continental volcanism in the Belcher Islands: implications for the evolution of the Circum Ungava Fold Belt. Canadian Journal of Earth Sciences, **31**: 1536–1549.

Müller, S.G., Krapež, B. Barley, M.E., and Fletcher, I.R. 2005. Giant iron-ore deposits of the Hamersley province related to the breakup of Paleoproterozoic Australia: new insights from in situ SHRIMP dating from mafic intrusions. Geology, **33**: 577–580.

Noble, S.R., and Lightfoot, P.C. 1992. U–Pb baddeleyite ages of the Kerns and Triangle Mountain intrusions, Nipissing Diabase, Ontario. Canadian Journal of Earth Sciences, **29**: 1424–1429.

Osmani, I.A. 1991. Proterozoic mafic dike swarms in the Superior Province of Ontario. *In* Geology of Ontario, Special Vol. 4, Part 1. Ontario Geological Survey, pp. 661–681.

Parrish, R.R. 1989. U–Pb geochronology of the Cape Smith Belt and the Sugluk block, northern Quebec. Geoscience Canada, **16**: 126–130.

Percival, J.A., and Peterman, Z.E. 1994. Rb–Sr biotite and whole-rock data from the Kapuskasing uplift and their bearing on the cooling and exhumation history. Canadian Journal of Earth Sciences, **31**: 1172–1181.

Percival, J.A., and West, G.F. 1994. The Kapuskasing uplift: a geological and geophysical synthesis. Canadian Journal of Earth Sciences, **31**: 1256–1286.

Percival, J.A., Mortensen, J.K., Stern, R.A., Card, K.D., and Bégin, N.J. 1992. Giant granulite terrane of northeastern Superior Province: the Ashuanipi complex and Minto block. Canadian Journal of Earth Sciences, **29**: 2287–2308.

Percival, J.A., Palmer, H.C., and Barnett, R.L. 1994. Quantitative estimates of emplacement level of post-metamorphic mafic dykes and subsequent erosion magnitude in the southern Kapuskasing uplift. Canadian Journal of Earth Sciences, **31**: 1218–1226.

Riller, U., Schwerdtner, W.M., Halls, H.C., and Card, K.D. 1999. Transpressive tectonism in the Penokean orogen, Canada. Consequences for Proterozoic crustal kinematics and continental fragmentation. Precambrian Research, **93**: 51–70.

Rollinson, H. 1993. Using geochemical data: evaluation, presentation, interpretation. Longman Group, Essex, England, 352 p.

Schwarz, E.J., and Fujiwara, Y. 1981. Paleomagnetism of the Circum-Ungava Fold belt II: Proterozoic rocks of the Richmond Gulf and Manitounuk island. *In* Proterozoic basins of Canada. *Edited by* F.H.A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 255–267.

Seguin, M.K., Sharma, K.N.M., Woussen, G., St. Seymour, K., and Symons, D.T.A. 1983. Paleomagnetism of Archean volcanics and Proterozoic dikes in La Grande Riviere area of Nouveau-Quebec. Canadian Journal of Earth Sciences, **20**: 339–343.

Shervais, J.W. 1982. Ti–V plots and the petrogenesis of modern and ophiolitic lavas. Earth and Planetary Science Letters, **59**: 101–118.

Symons, D.T.A., Quick, A.W., and Stupavsky, M. 1981. Magnetic and paleomagnetic characteristics of the Archean iron formation and host rocks at the Adams Mine, Ontario. Ontario Geological Survey, Miscellaneous Paper, 98, pp. 298–307.

Symons, D.T.A., Lewchuk, M.T., Dunlop, D.J., Constanzo-Alvarez, V., Halls, H.C., Bates, M.P., Palmer, H.C., and Vandall, T.A. 1994. Synopsis of paleomagnetic studies in the Kapuskasing Structural Zone. Canadian Journal of Earth Sciences, **31**: 1206–1217.

Thompson, R.N., Morrison, M.A., Dickin, A.P., and Hendry, G.L. 1983. Continental flood basalts … arachnids rule OK? *In* Continental basalts and mantle xenoliths. *Edited by* C.J. Hawkesworth and M.J. Norry. Shiva Publishing Ltd., Cheshire, UK, pp. 158–185.

Vuollo, J., and Huhma, H. 2005. Paleoproterozoic mafic dykes in NE Finland. *In* Precambrian geology of Finland—key to the evolution of the Fennoscandian Shield. *Edited by* M. Lehtinen, P.A. Nurmi, and O.T. Rämö. Elsevier, Amsterdam, The Netherlands, pp. 193–235.

West, G.F., and Ernst, R.E. 1991. Evidence from aeromagnetics on the configuration of Matachewan dykes and the tectonic evolution of the Kapuskasing structural zone, Ontario, Canada. Canadian Journal of Earth Sciences, **28**: 1797–1811.

York, D. 1969. Least-squares fitting of a straight line with correlated errors. Earth and Planetary Science Letters, **5**: 320–324.

© 2007 NRC Canada